

Road map!

- Module 1- Demystifying Timeseries Data and Modeling
- Module 2- Setting up Deep Forecasting Environment
- Module 3- Exponential Smoothing
- Module 4- ARIMA models
- Module 5- Machine Learning for Time series Forecasting
- Module 6- Deep Neural Networks
- Module 7- Deep Sequence Modeling (RNN, LSTM)
- Module 8- Prophet and Neural Prophet



Pedram Jahangiry

Module 1

Demystifying Timeseries Data and Modeling



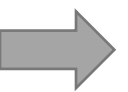
TS Basics
Modeling
Forecasting



???



Pedram Jahangiry



Key topics

1. Timeseries Basics

- What is sequence data?
- Time series tasks
- Trend, Seasonality, Residual
- ACF and PACF
- Stationarity

3. Modeling

- White Noise
- Econometrics
- Machine Learning
- Deep Learning

2. Forecasting strategies

- One-step, Multi-step, Multi-output
- Univariate vs Multivariate
- Benchmarks

4. Can we beat wall street?

- Is stock price Random Walk? (EMH)
- Does Uni-variate Model work?
- Can Multi-variate Model help?
- So What?





Module 1- Part 1

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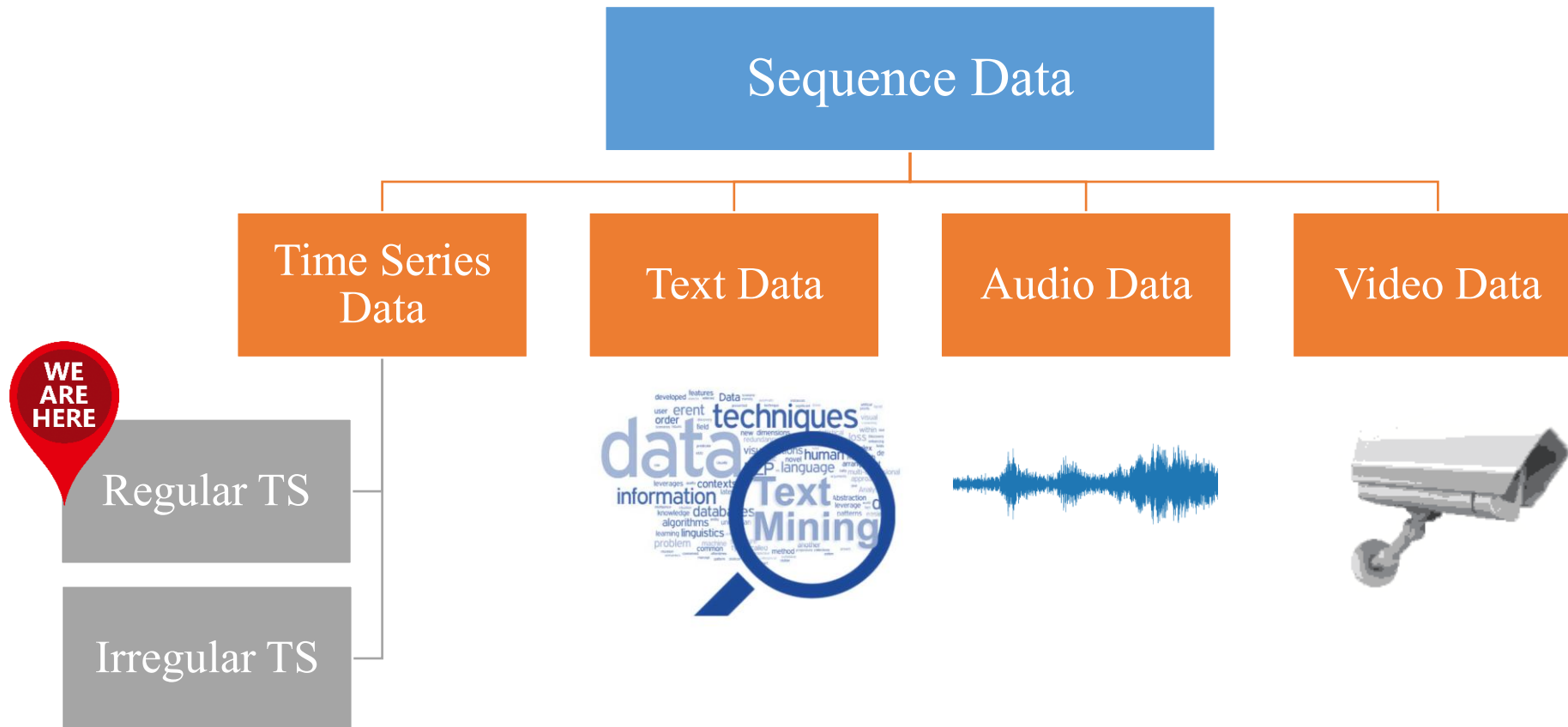
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What is Sequence Data?

Sequence data refers to any data that has a specific **order** or sequence to it!



Time series Tasks

TS tasks

Forecasting

Classification

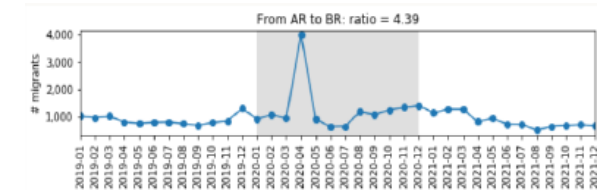
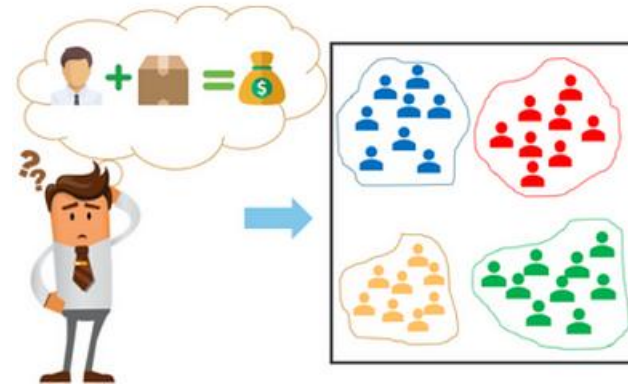
Clustering

Anomaly/Event Detection

Qualitative

Quantitative

WE
ARE
HERE



Hey Siri



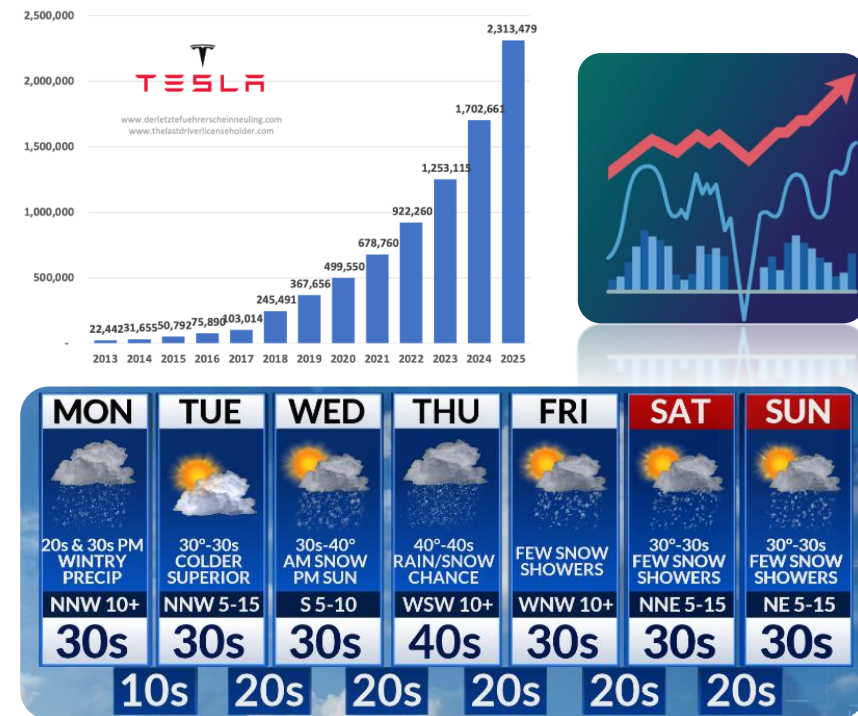
Assumption: past patterns will **continue**



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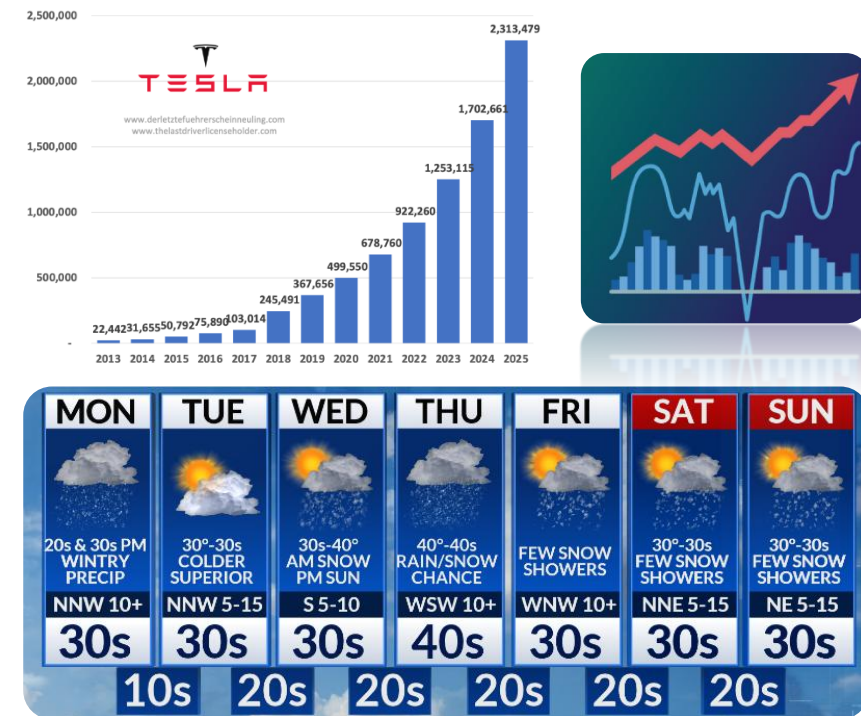
More Forecasting examples

- Which of the following examples are easier to forecast?
 1. Time of sunset this day next month
 2. Apple stock price tomorrow
 3. Airline ticket demand/price next year
 4. New car model sales in the first quarter
 5. Monthly rainfall in Utah next winter
- What impacts Forecastability?
 - Data Availability
 - How **similar** the future is to the past!
 - Good understanding of the **underlying factors**

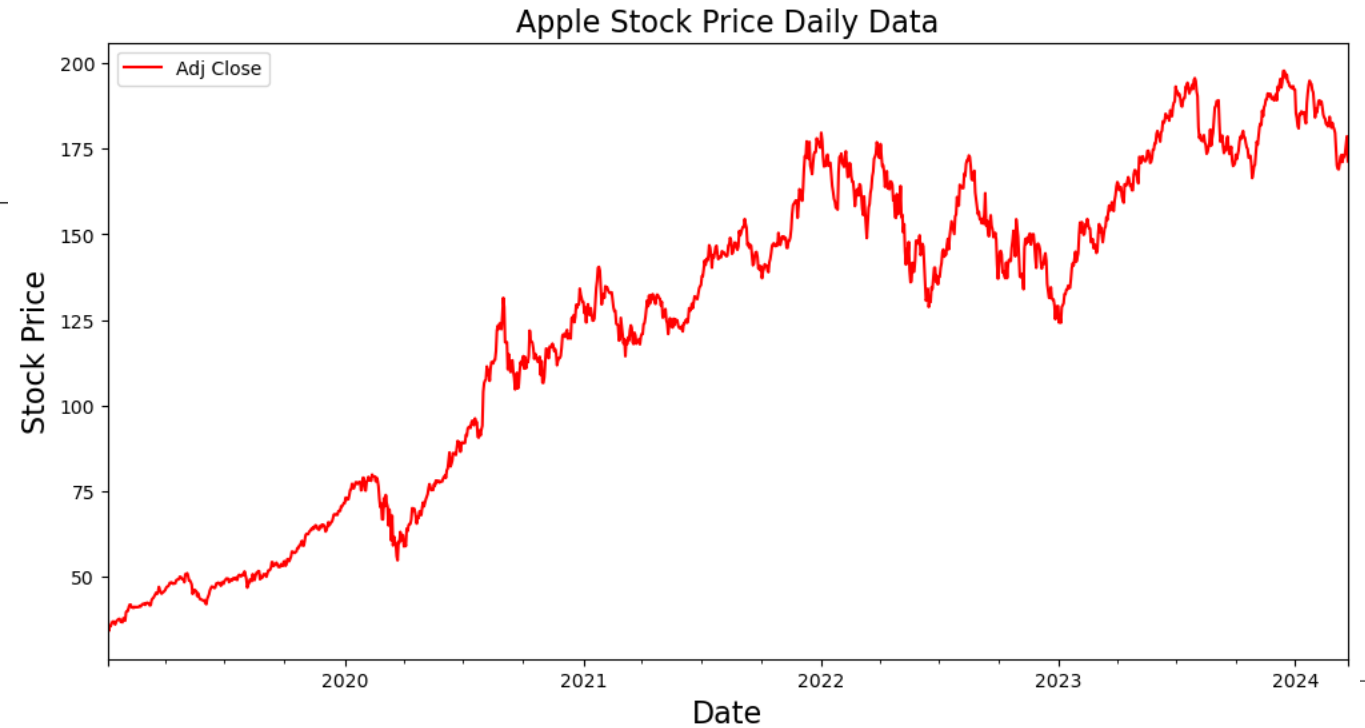
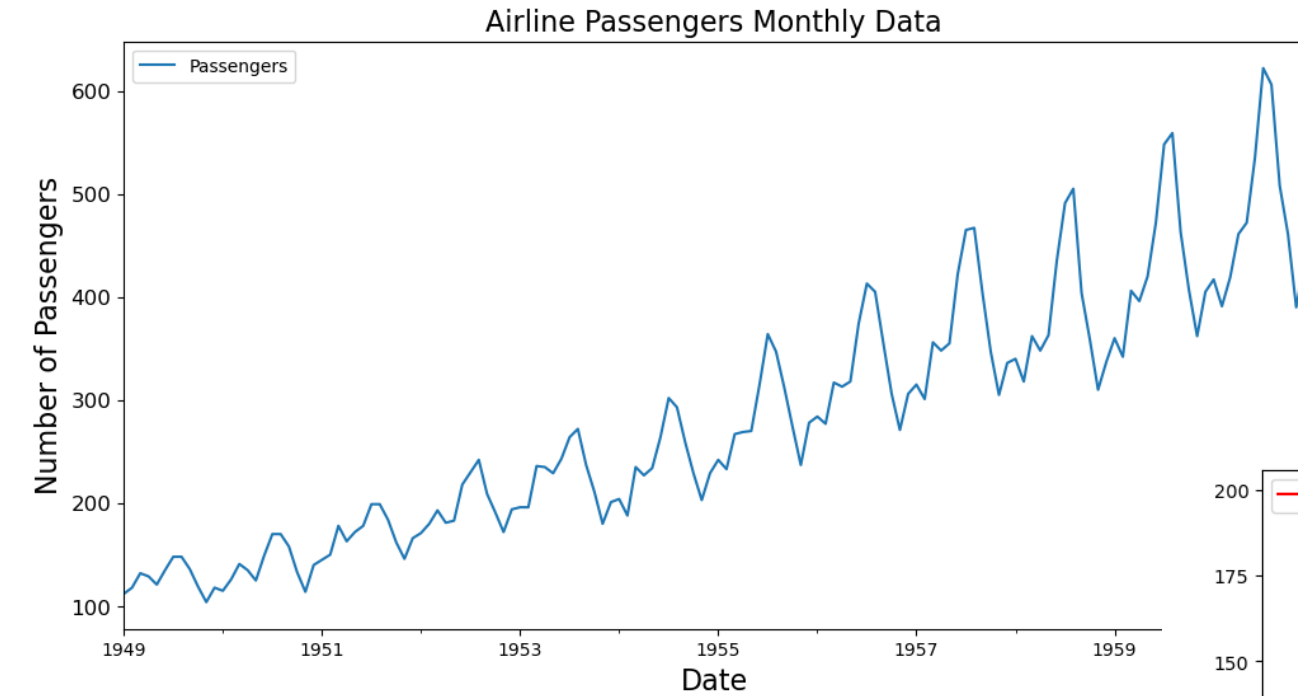


More Forecasting examples

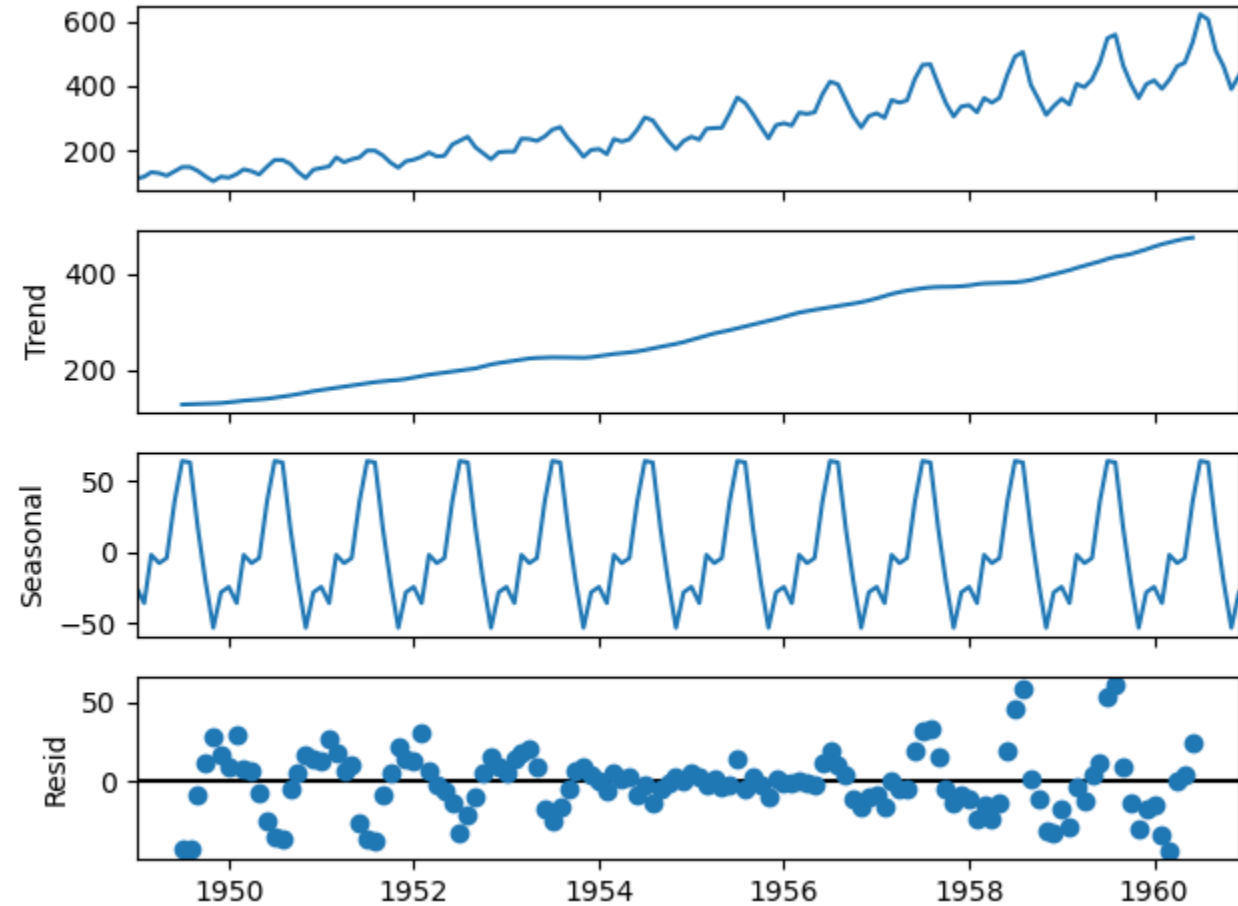
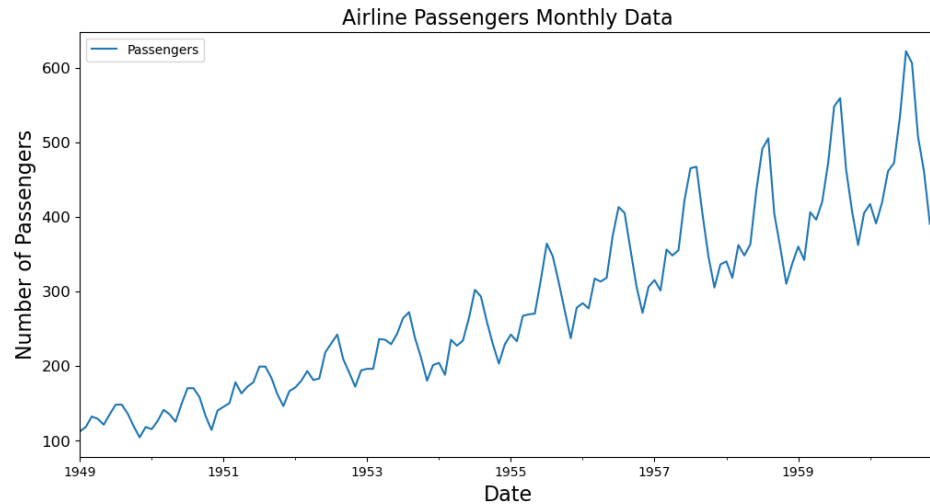
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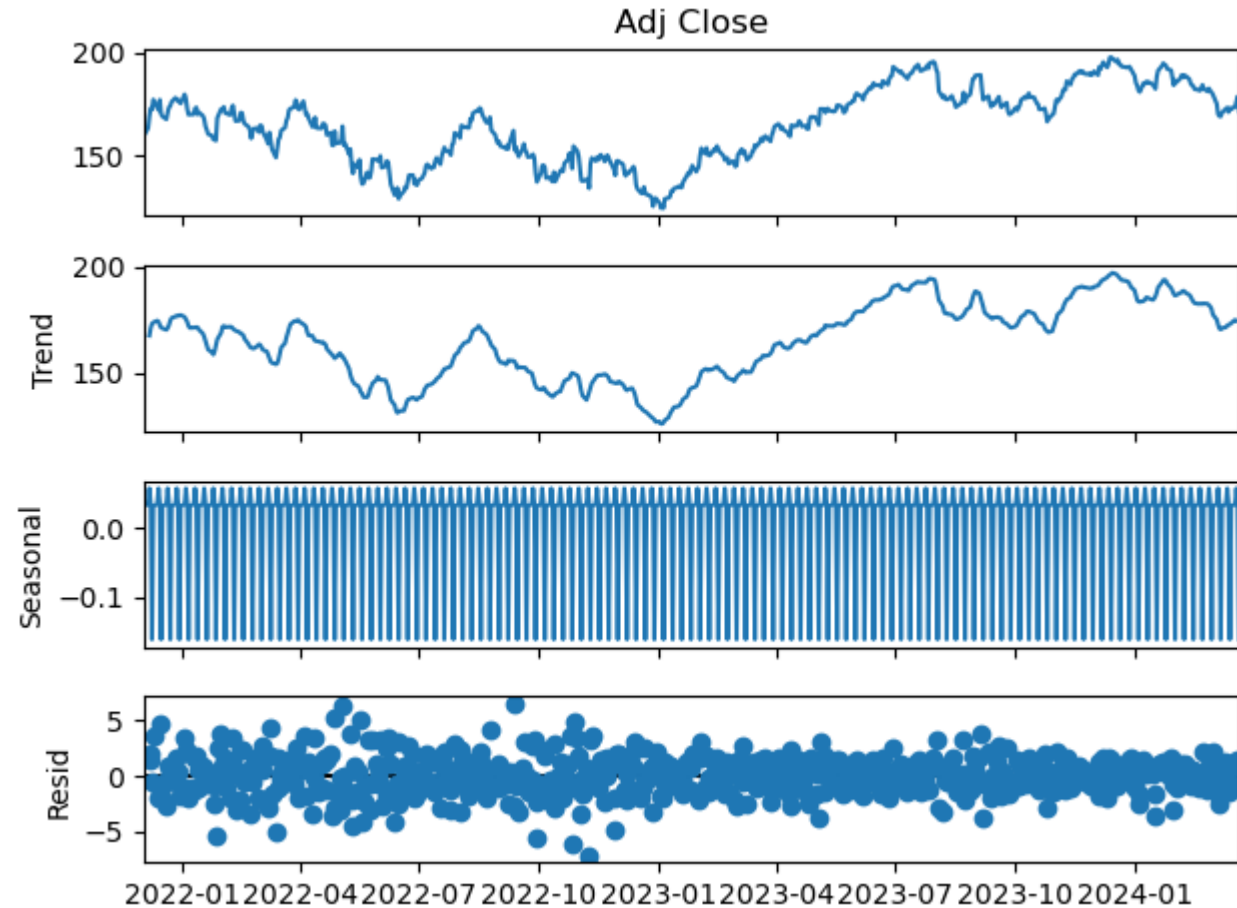
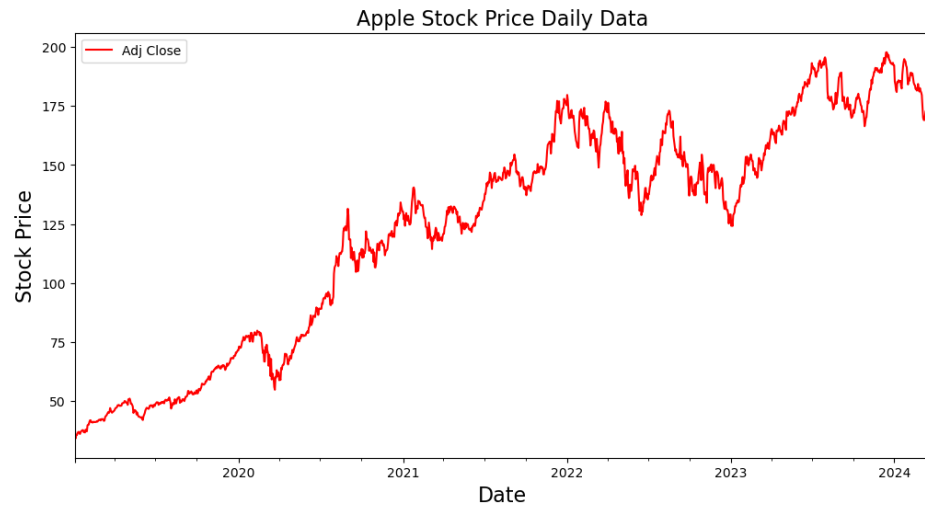
Quantitative Forecasting



TS Decomposition: Trend, Seasonality, Residual

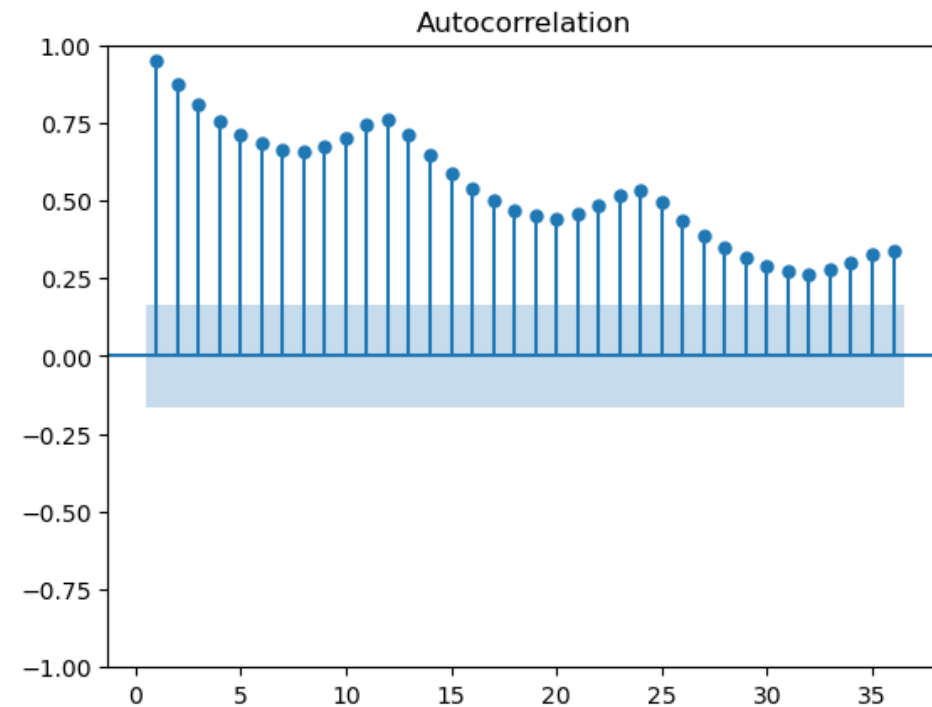
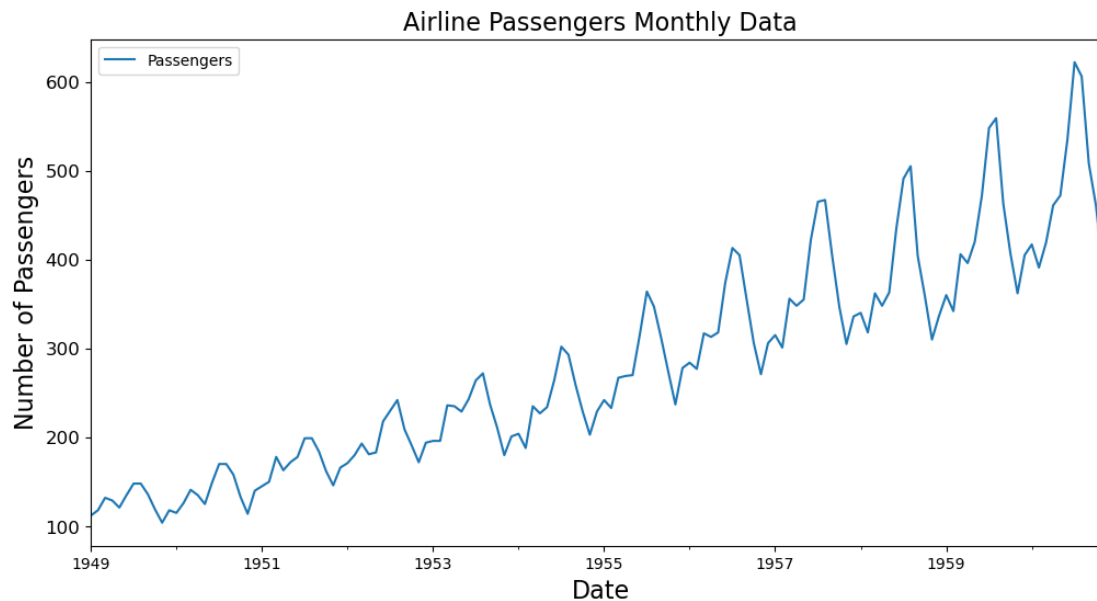


TS Decomposition: Trend, Seasonality, Residual



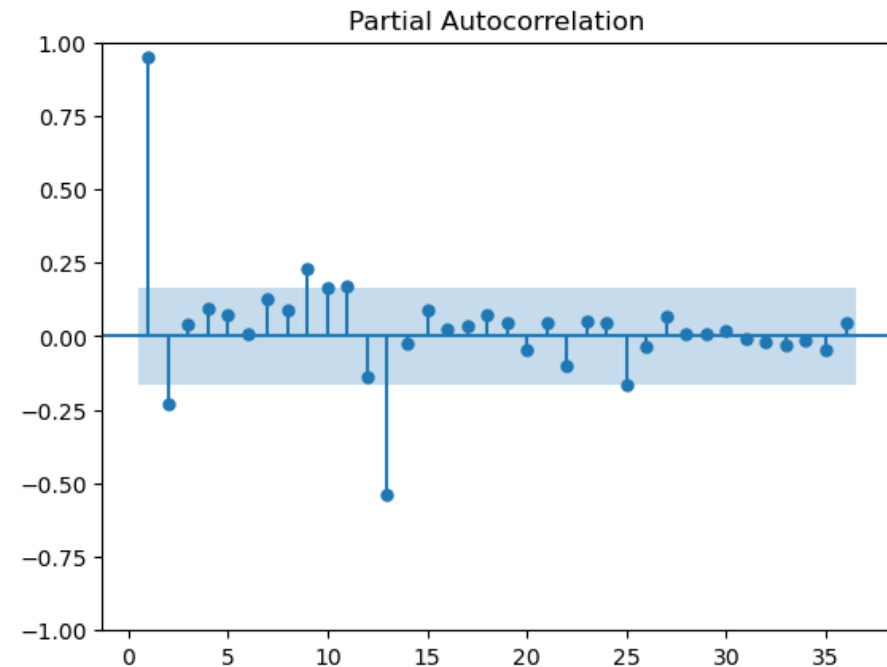
ACF: Autocorrelation Function

- **ACF** is a statistical tool that can be used to measure the autocorrelation of a time series.
- **Autocorrelation (Serial correlation)** is a measure of the correlation between a time series and a lagged version of itself.
- ACF shows the **degree to which** the past values **are predictive** of future values.

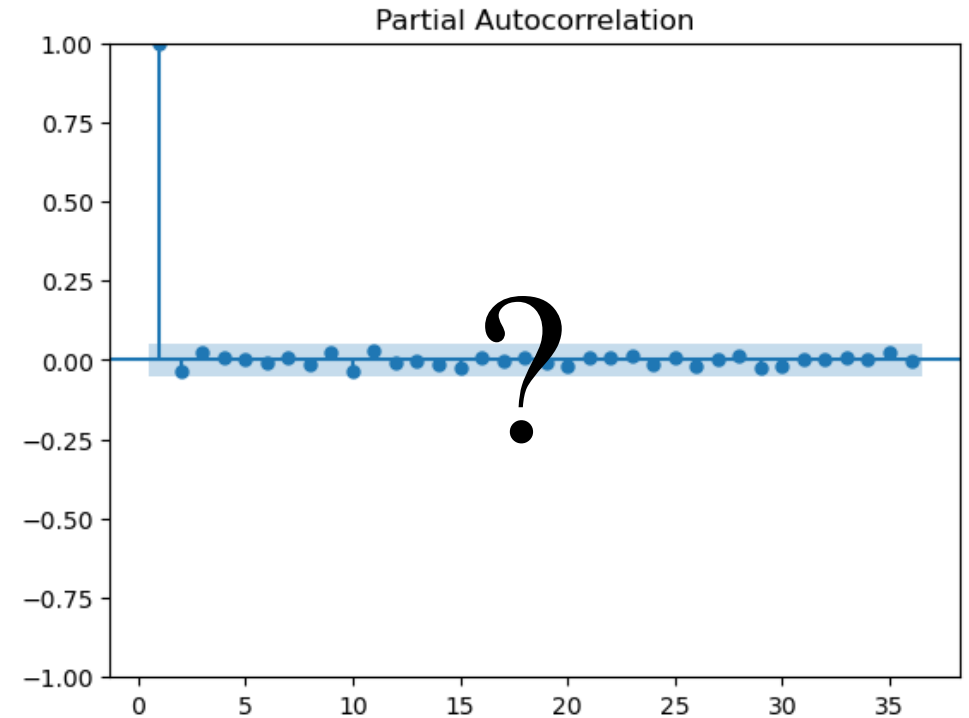
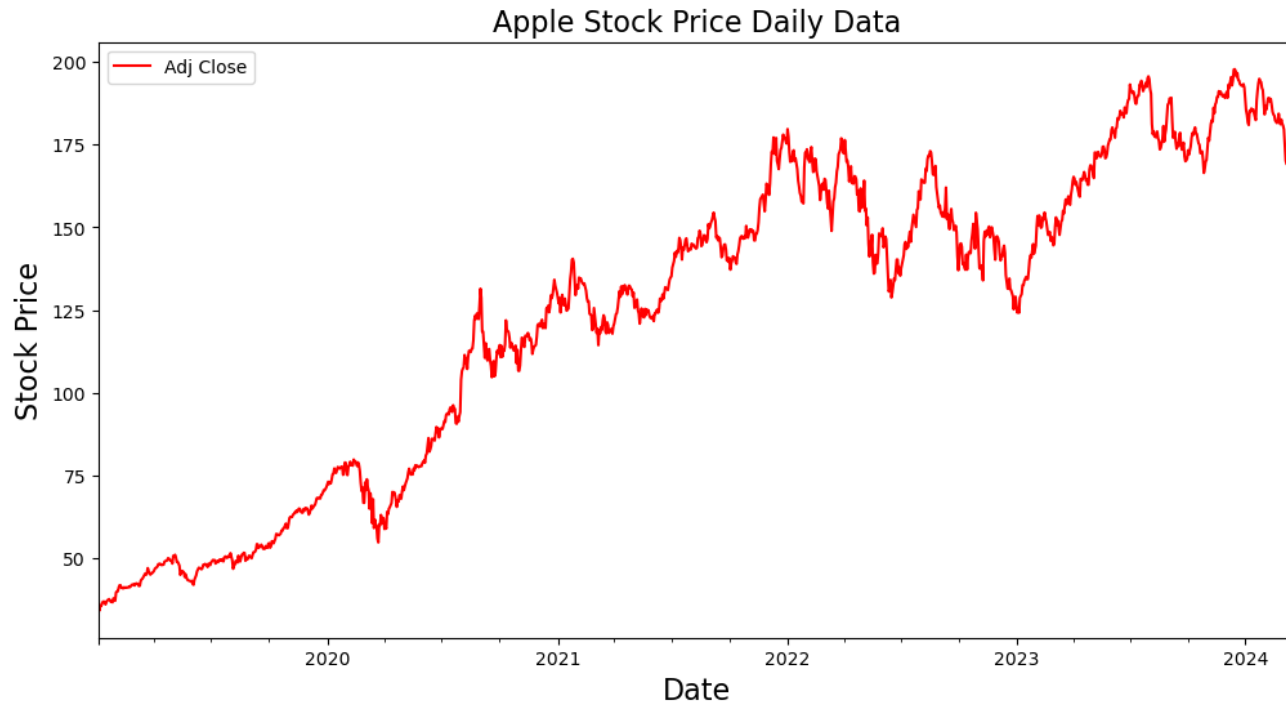


PACF: Partial Autocorrelation Function

- PACF measures the direct correlation between time series and its lag, **controlling for intermediate lags**.
- y_t and y_{t-2} might be correlated, simply because they are both connected to y_{t-1}

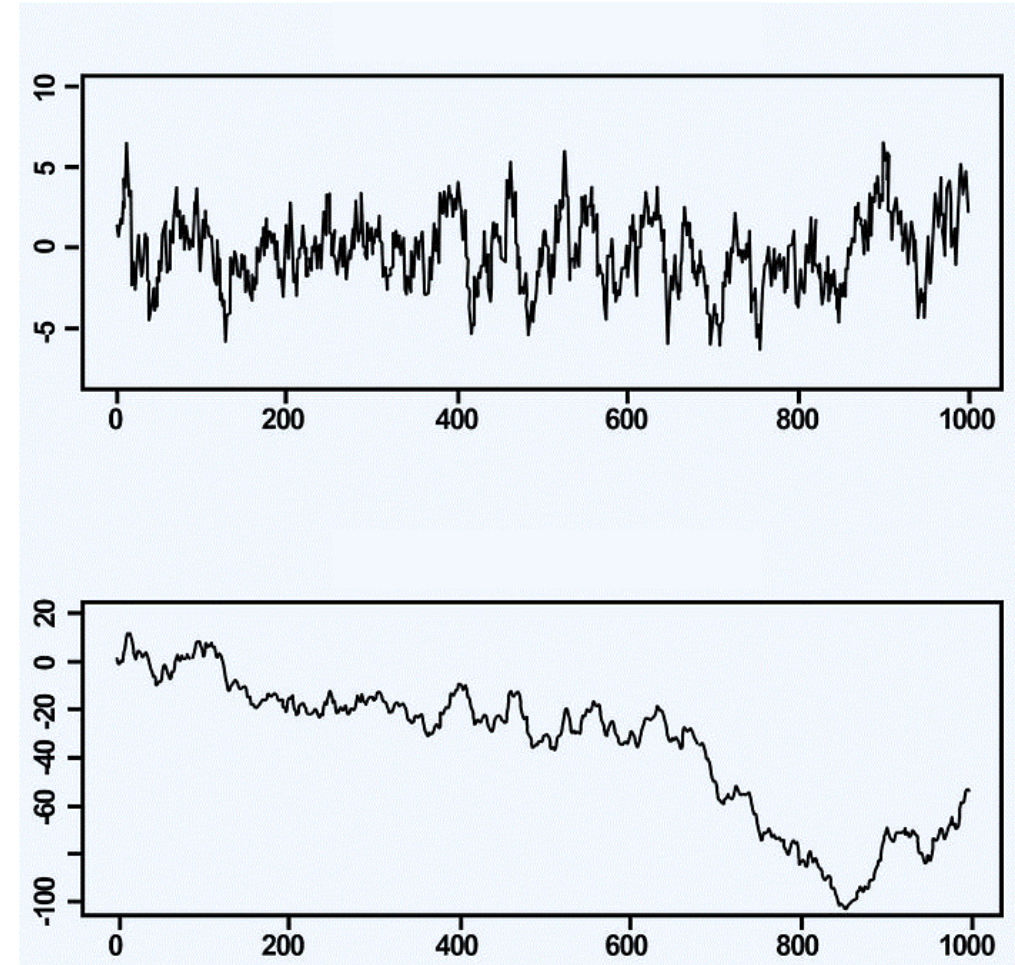


PACF for AAPL Stock Price?



Stationarity

- Stationary vs Non-Stationary Data. What makes a data set **Stationary**?
- In a stationary timeseries, the statistical properties **do not depend on the time**
- **Predictability (long-horizon)**: Stationary time series are easier to forecast because you can assume that future statistical properties will not change.
- **This doesn't mean we cannot forecast non-stationary data!**



Differencing

- Differencing: Computing the difference between consecutive observations.
- Differencing helps to stabilize the mean of a time series by removing changes in the level

Original data 1st differenced 2nd differenced

Time1	10
Time2	12
Time3	8
Time4	14
Time5	7

Time1	NA
Time2	2
Time3	-4
Time4	6
Time5	-7

Time1	NA
Time2	NA
Time3	-6
Time4	10
Time5	-13





Module 1- Part 1

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3. Modeling

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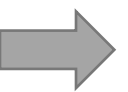
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- One-step, Multi-step, Multi-output
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4. Can we beat wall street?

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Module 1- Part 2

1. Timeseries Basics

- What is sequence data?
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3. Modeling

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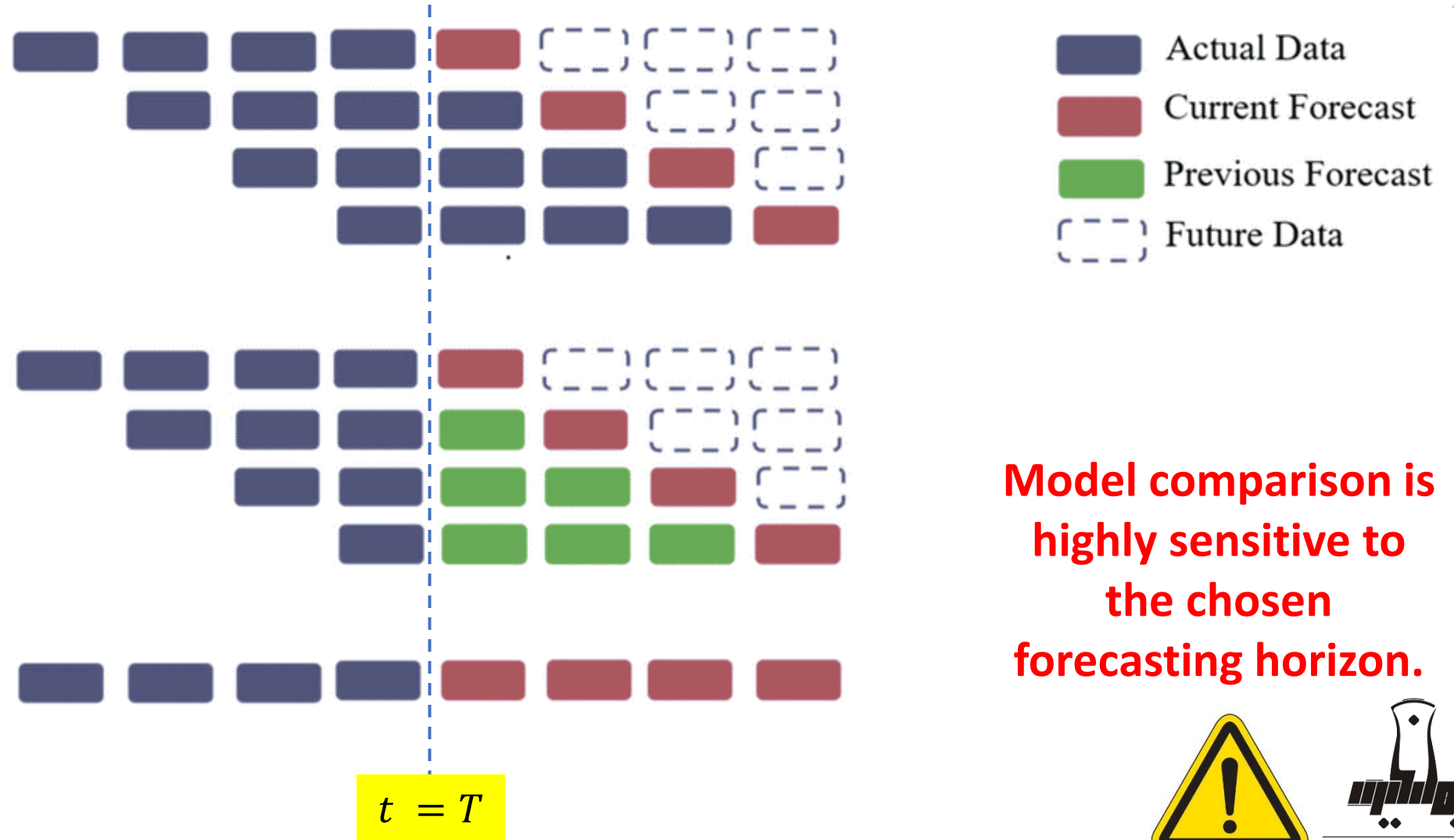
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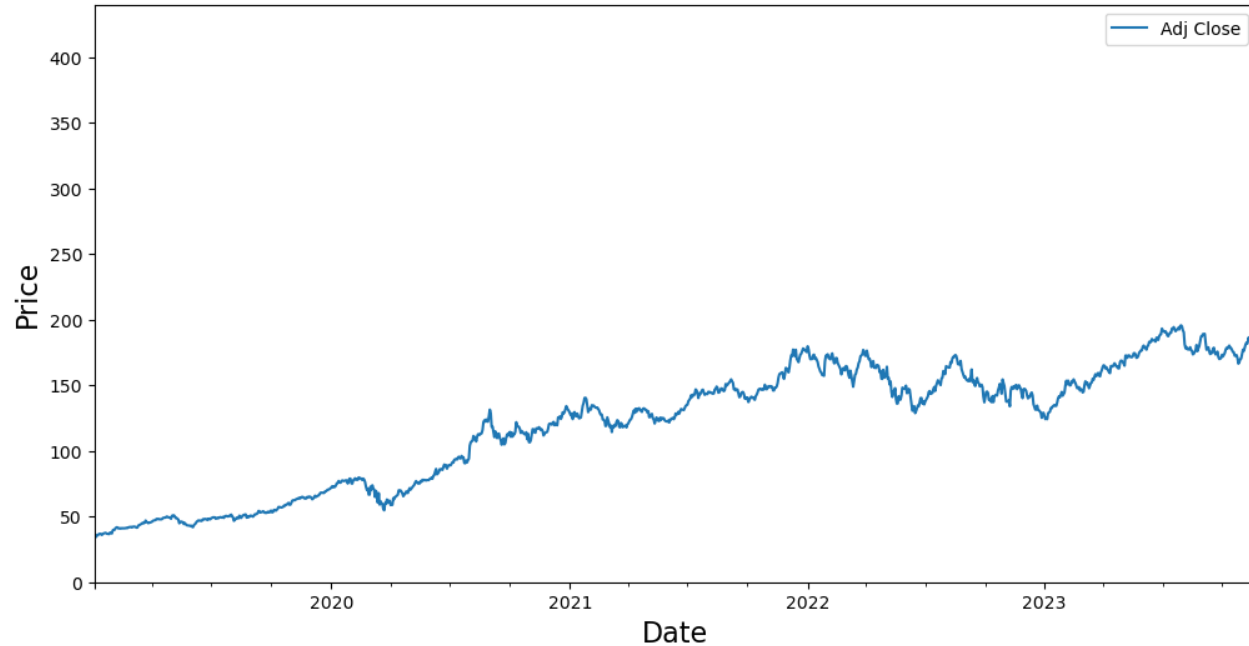


One-step, Multi-step, Multi-output Forecasts



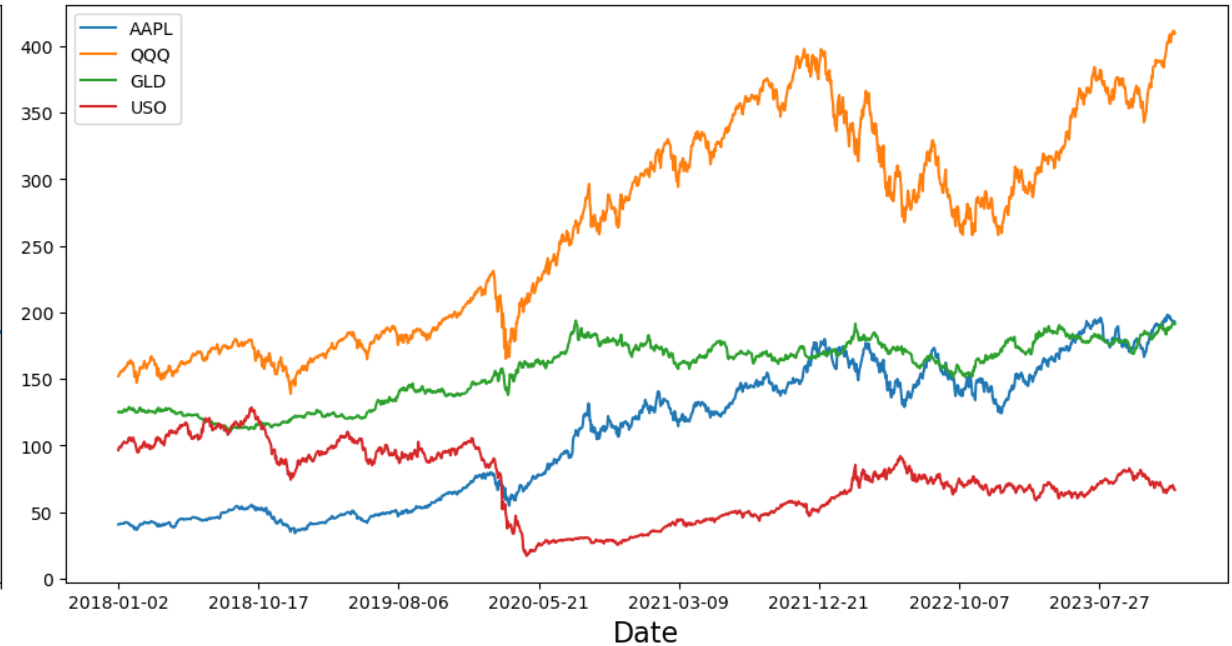
Univariate vs Multivariate

Univariate Time Series



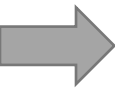
- Using one variable's history to forecast that same variable

Multivariate time series



- Uses multiple variables and their relationships to forecast future values
- **Multi-step forecasts require future predictor values, forecasts, or scenarios.**
- Less stable in LR due to increased forecast error.





Forecasting Benchmarks

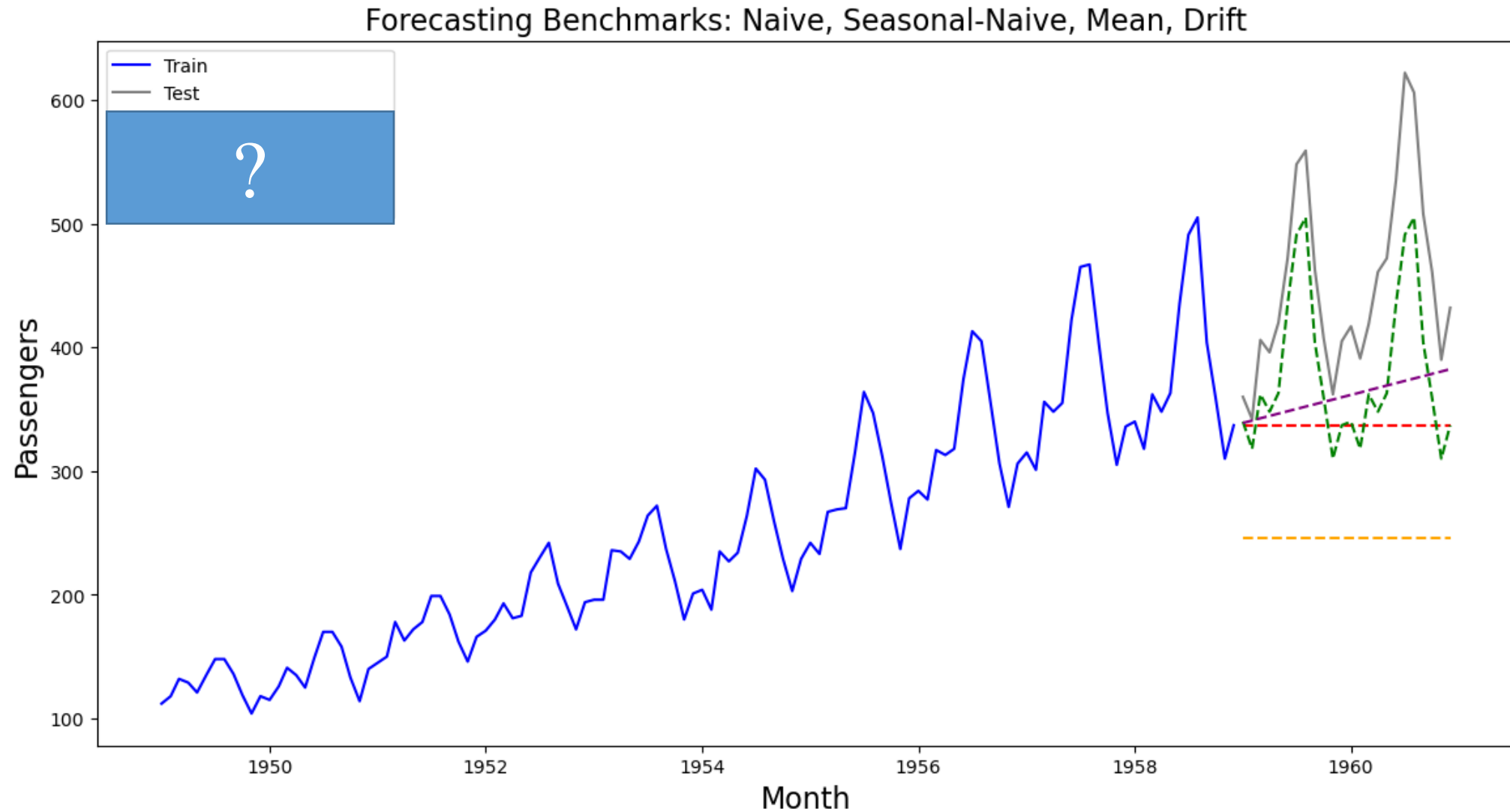
- Some forecasting methods are **extremely simple** and **surprisingly effective**.

- Mean** method: Predicts all future values to be the average of the historical data.
- Naïve** method: Predicts the next value to be the current value
- Seasonal Naïve**: Predicts the next value by using the same value from the previous season
- Drift** method: Predicts future values by extending a linear trend fitted to the historical data.

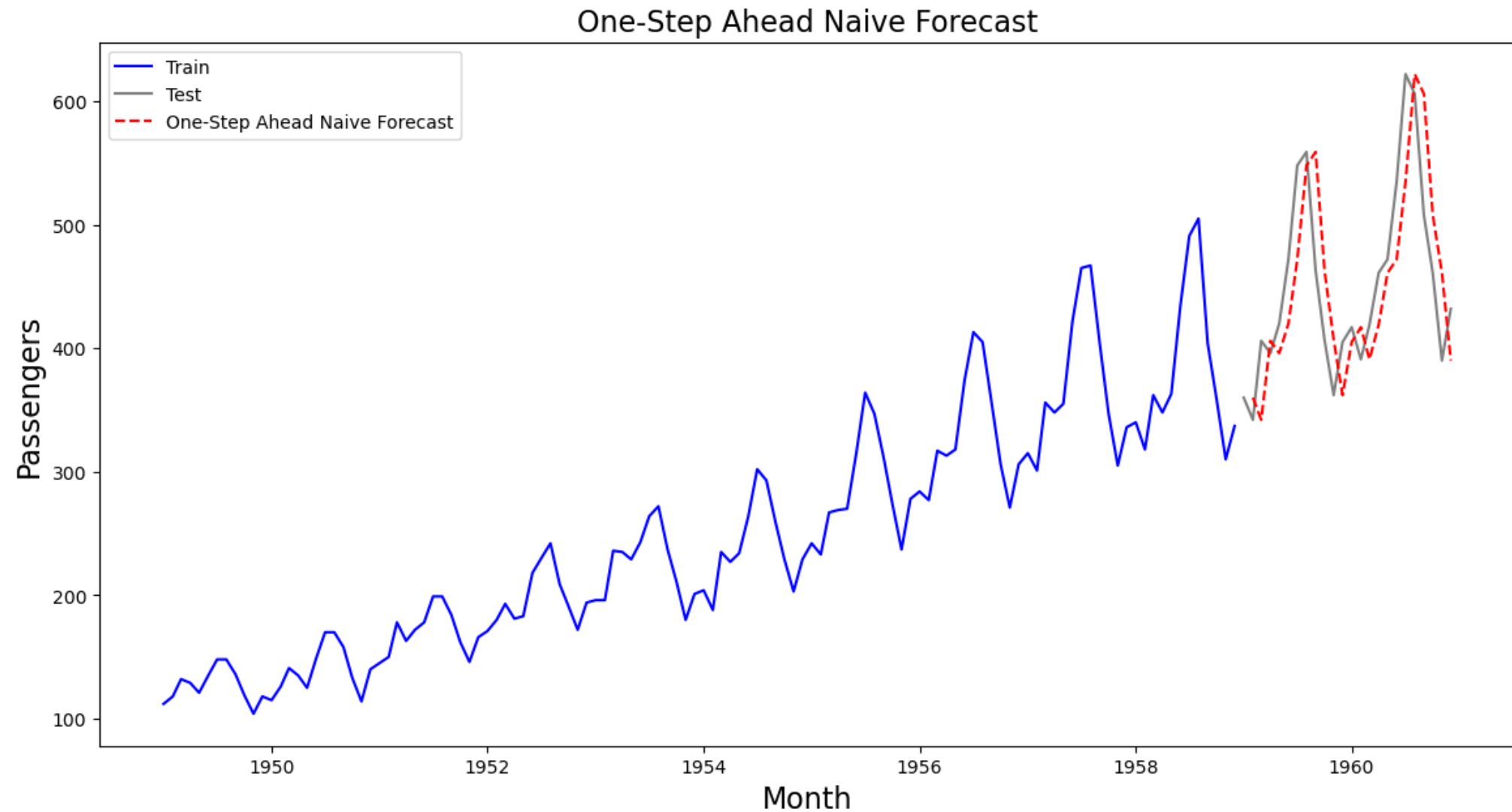
- These methods will serve as benchmarks rather than the method of choice.
- If our model cannot beat the benchmark, **it is not** worth considering!!!



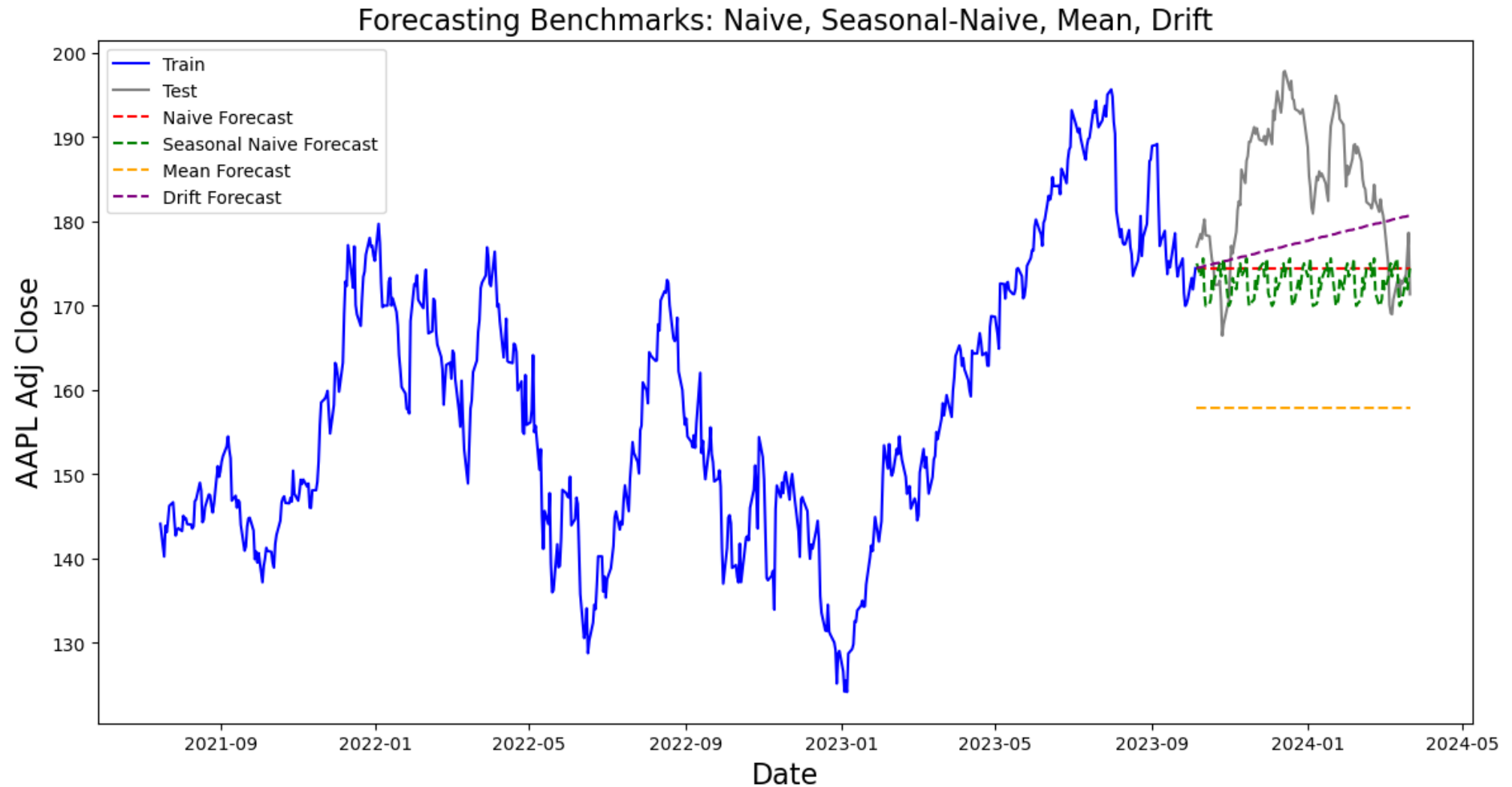
Forecasting Benchmarks (Multi-Horizon)



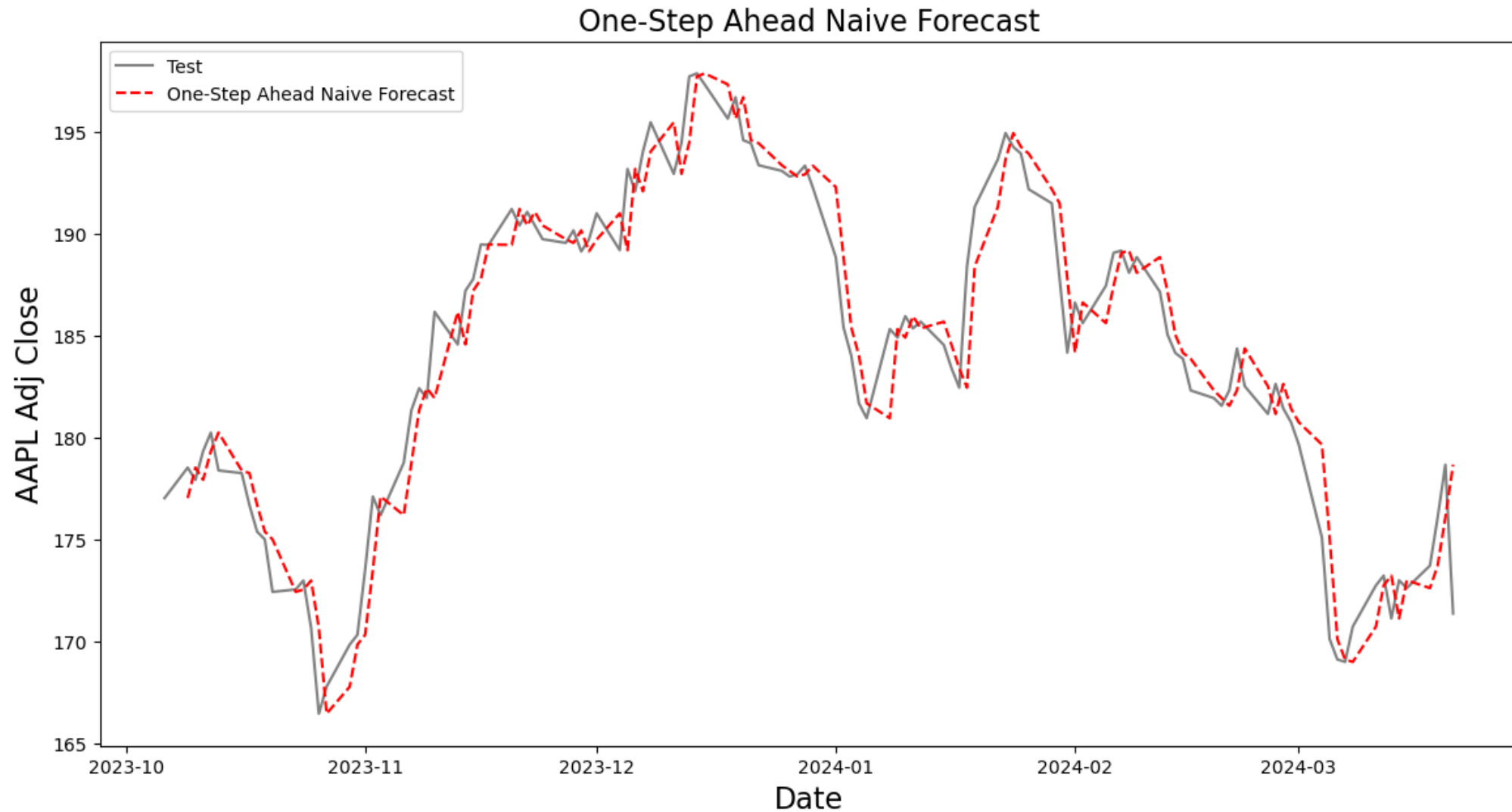
Forecasting Benchmarks (One-step)

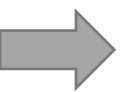


Forecasting Benchmarks



Forecasting Benchmarks





Module 1- Part 3

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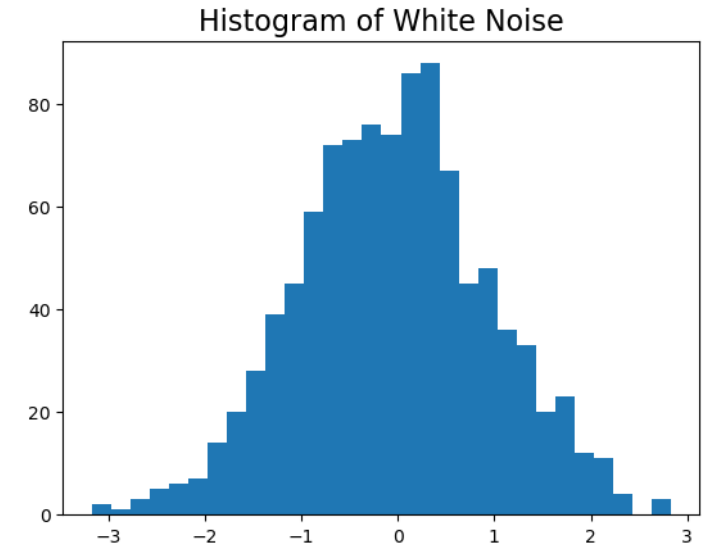
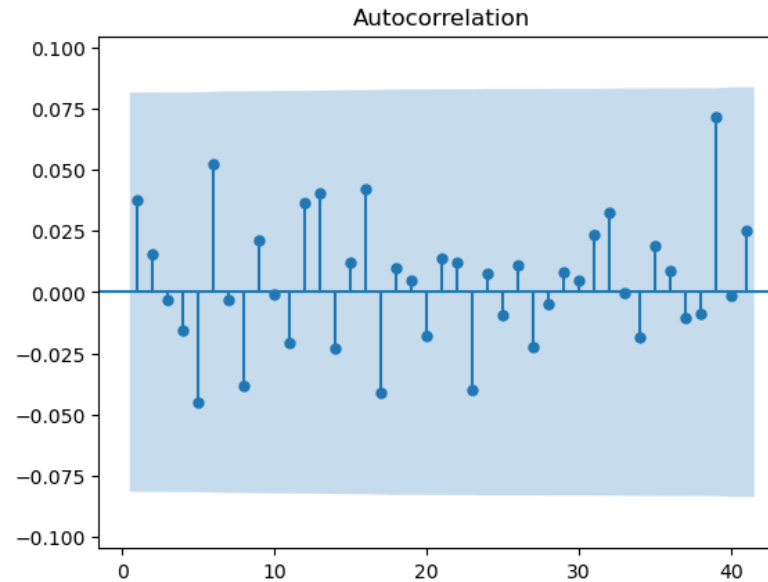
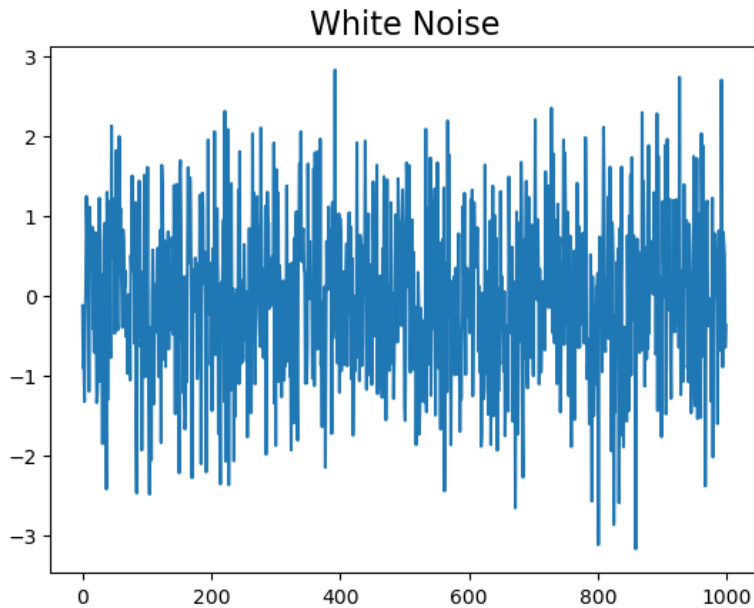
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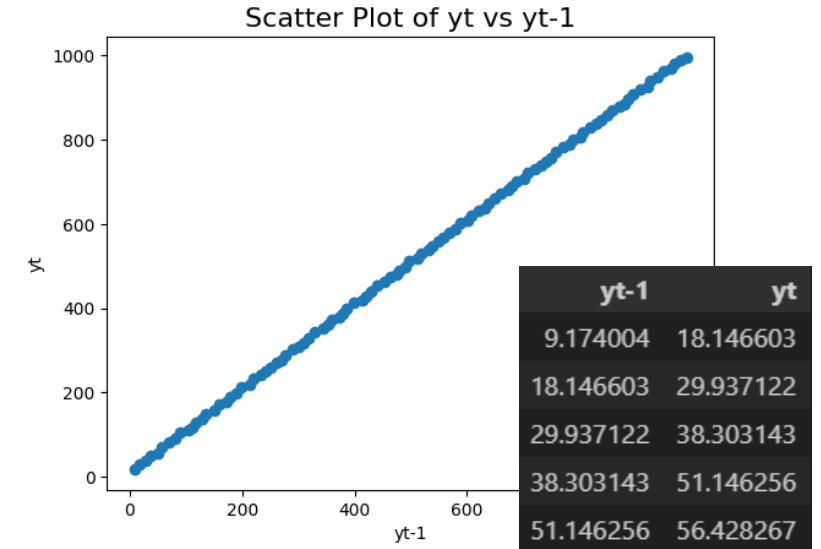
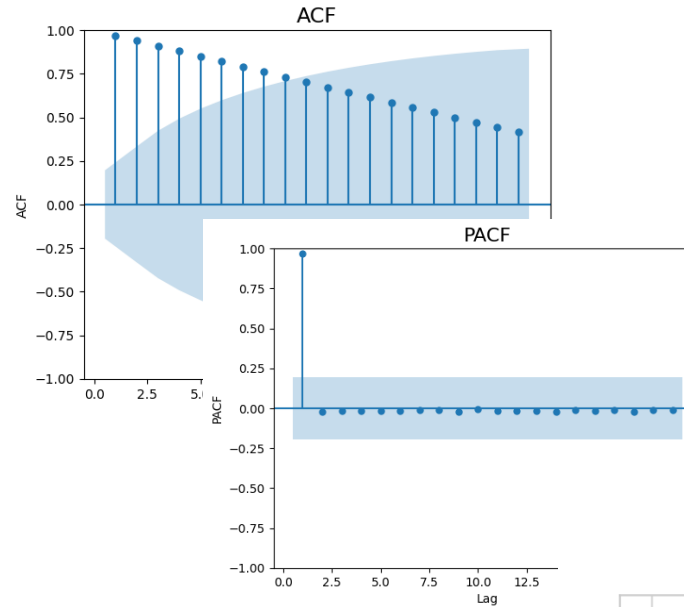
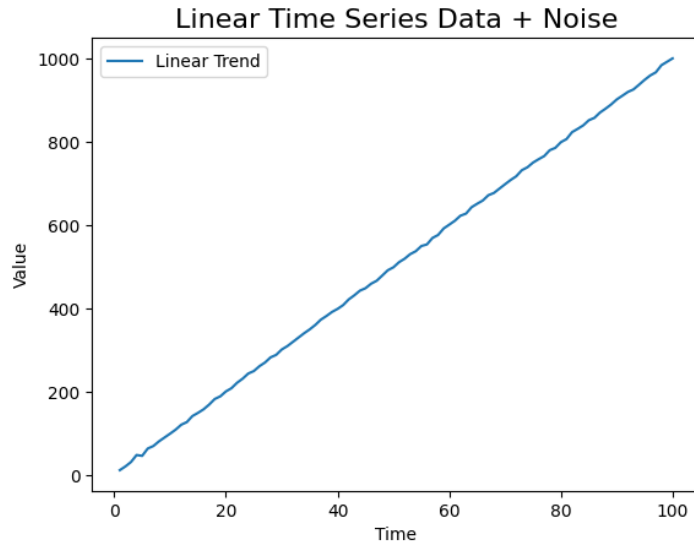


The Importance of White Noise in Modeling

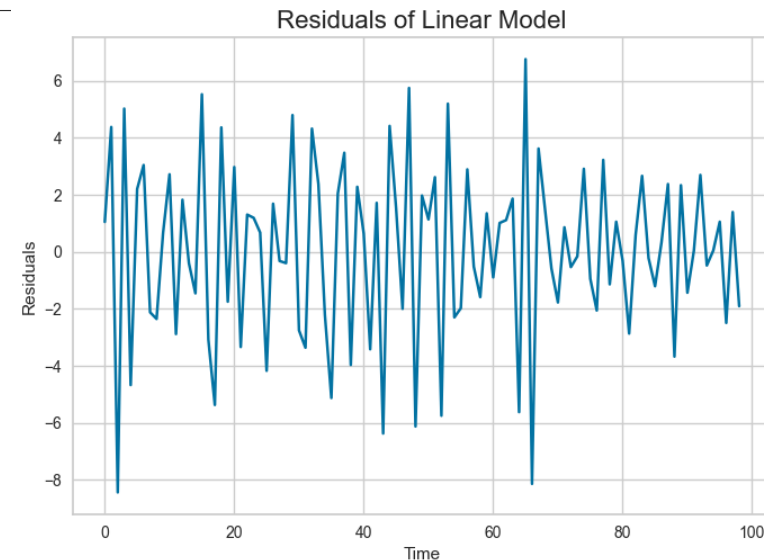
- White noise refers to a random process with:
- **Zero mean**, **Constant variance** and **No autocorrelation**.
- Role in timeseries modeling: **Diagnostic tool!** Indicator of a good fit!



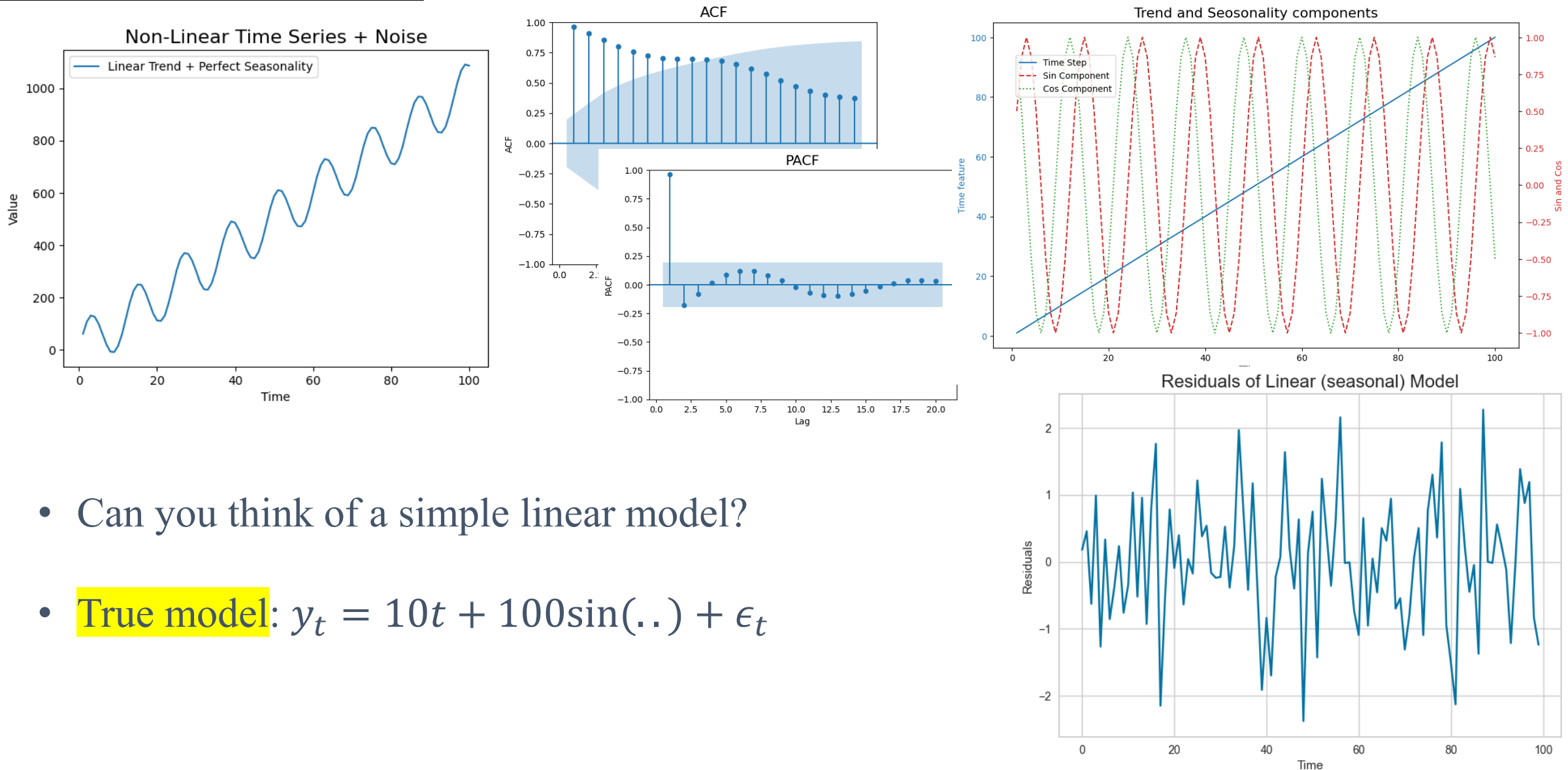
Linear vs Non-Linear (data or model?)



- Can you think of a simple linear model?
- True model:** $y_t = 10 + y_{t-1} + \epsilon_t$

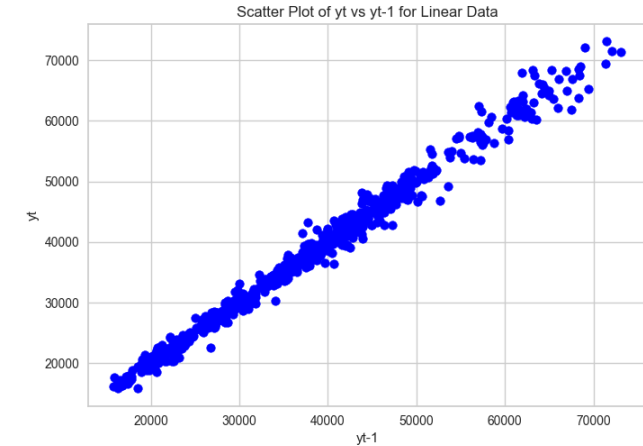
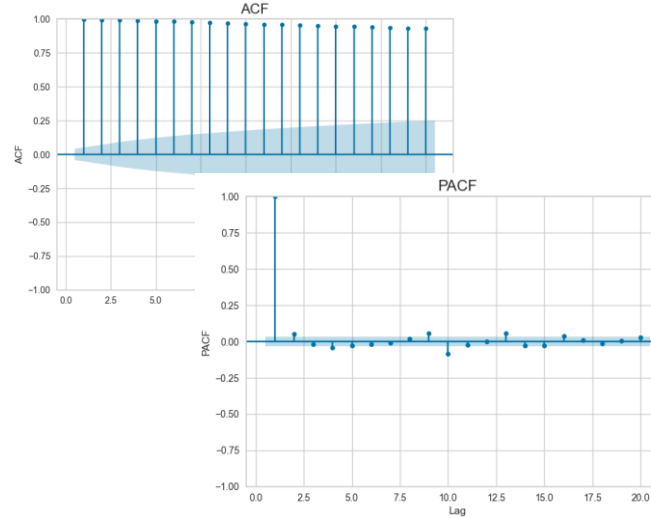
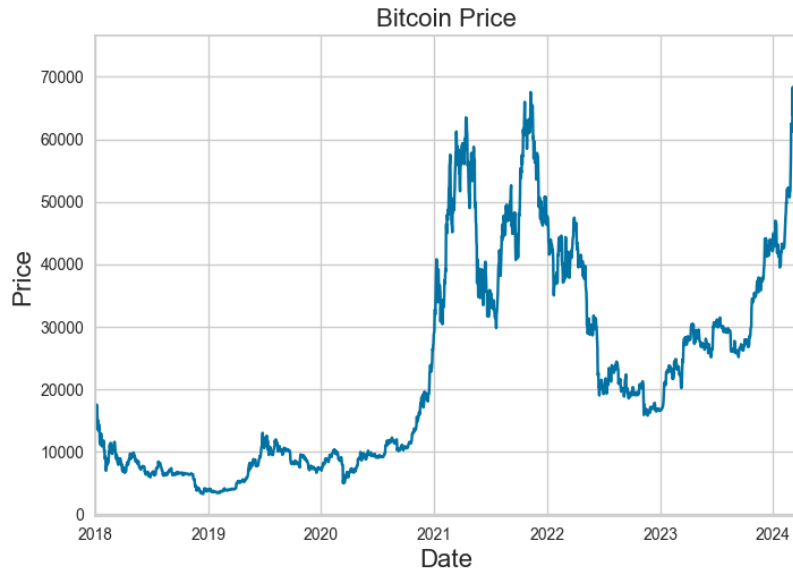


Linear vs Non-Linear (data or model?)

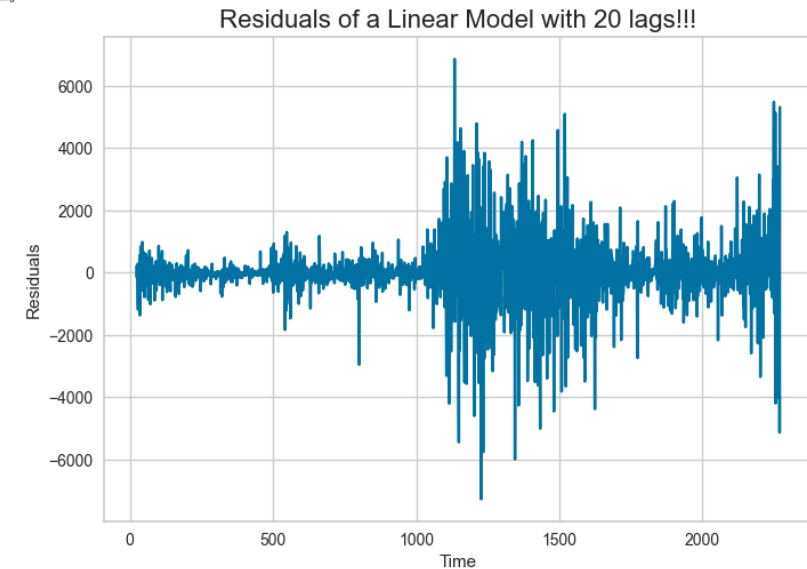


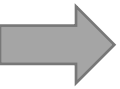
- Can you think of a simple linear model?
- **True model:** $y_t = 10t + 100\sin(\cdot) + \epsilon_t$

Linear vs Non-Linear (data or model?)



- Can you think of a simple linear model?
- True model: $y_t = I \text{ wish :)}$





Basic steps in a forecasting task

Step 1: Problem definition

- Forecasting **type** and **horizon** (one-step, multi-step, multi-output forecasts), ...

Step 2: Data Collection

- Time horizon, structural changes, **data type**, ...

Step 3: Exploratory Analysis

- Trend, seasonality, outliers, ...

Step 4: Model Selection and Training

- **Traditional** vs **machine learning** vs **deep learning**

Step 5: Model Evaluation and Comparison

- MSE, RMSE, MAE, MAPE, ...



→ Modeling, a simple example

- Quantifying airline passengers!
- Let's build a model:
 - Cross sectional vs timeseries?
 - Univariate vs Multivariate?

$$\text{air passengers} = \beta_0 + \beta_1 \text{gdp} + \beta_2 \text{unemp} + \beta_3 \text{saving} + \dots + u$$

- ✓ Can you interpret this model?
- ✓ Can you make predictions using your model?
- ✓ Can you make forecasts into the future? What are the challenges?



→ Modeling, a simple example (cont'd)

- Focusing on Forecasting for one country:

$$\text{air passengers}_t = \beta_0 + \beta_1 \text{gdp}_t + \beta_2 \text{unemp}_t + \beta_3 \text{saving}_t + \dots + u_t$$

- This **explanatory** model is contemporaneous (static). How can we make it a **dynamic forecasting model**?

$$\text{air passengers}_{t+1} = \beta_0 + \beta_1 \text{gdp}_t + \beta_2 \text{unemp}_t + \beta_3 \text{saving}_t + \dots + u_t$$

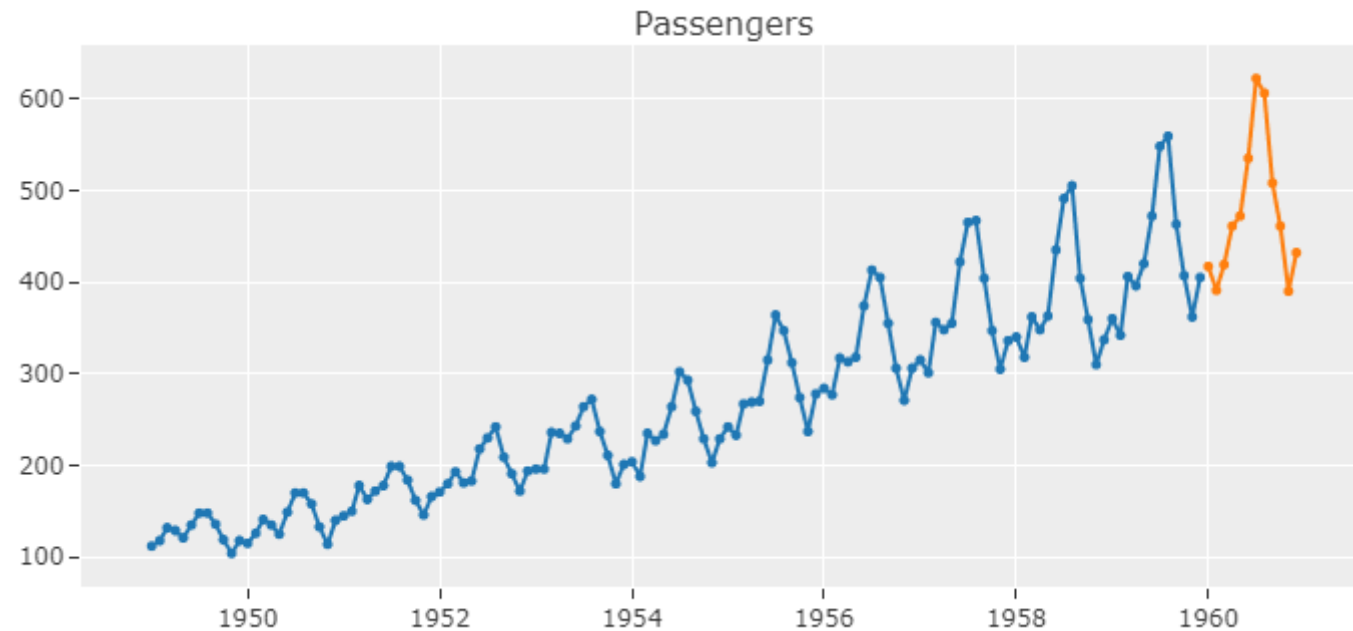
- Assumption**: everything is **reflected** in passengers' number already! So, why not →

$$\text{passengers}_{t+1} = f(\text{passengers}_t, \text{passengers}_{t-1}, \dots)$$



→ Modeling, a simple example (cont'd)

$$\text{passengers}_{t+1} = f(\text{passengers}_t, \text{passengers}_{t-1}, \dots)$$



- Goal: Identify **existing patterns** in data to project **future patterns**!

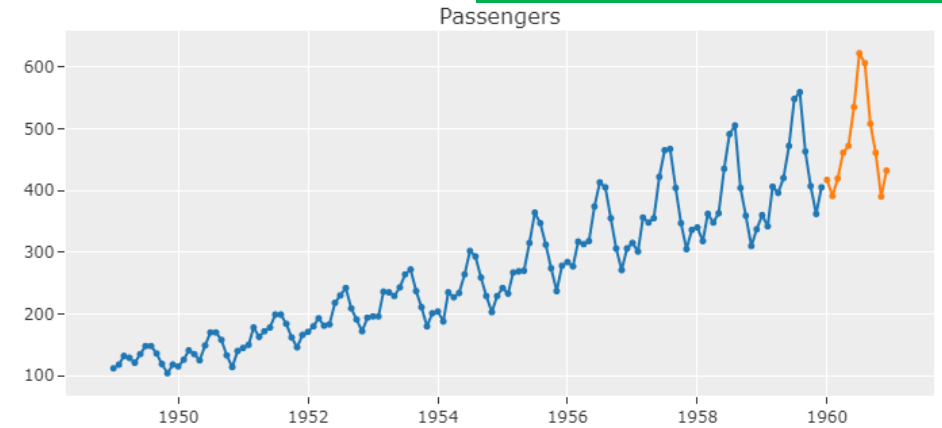
Why do we need ML/DL?

$$passengers_{t+1} = f(passengers_t, passengers_{t-1}, \dots)$$

- How do we identify the functional form of $f(\cdot)$?
- How about this:

$$passengers_{t+1} = c + \phi_1 passengers_t + \phi_2 passengers_{t-1} + \dots + \epsilon_t$$

- What if we are looking for **more complex** relationships?
- Or a **non-parametric** approach?



→ Predictive modeling vs forecasting?

$$passengers = \beta_0 + \beta_1 gdp + \beta_2 unemp + \beta_3 saving + \dots + u$$

	Predictive Modeling	Forecasting
Definition	Creating models to analyze data and predict an outcome.	Making informed predictions about future events based on past data.
Focus	Any point in time (past, present, future)	Primarily focused on future outcomes.
Variables	Multivariate	Typically, univariate

- Forecasting is a subset of predictive modeling.
- We can use econometrics, ML or DL for predictive modeling.

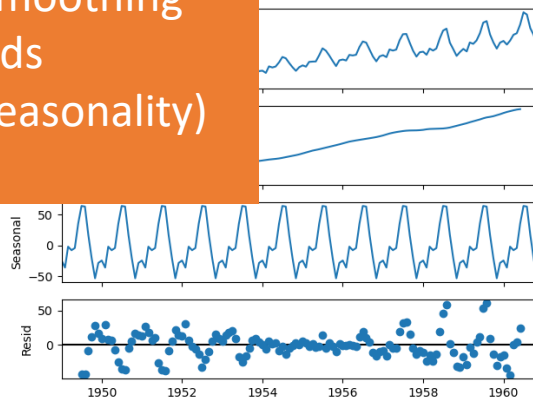


Econometrics

- **Decomposition-Based**: Focuses on modeling through **decomposition** and **autocorrelations**.
- **Parametric Structure**: Defined mathematical form; **interpretable** components.
- **Statistical Tools**: Confidence intervals, hypothesis testing for insights.

ETS

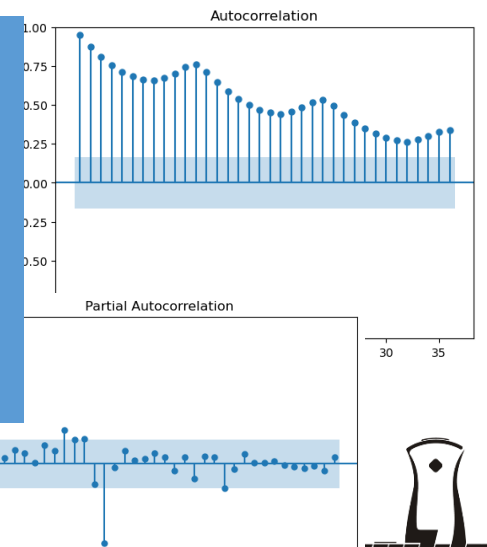
Exponential Smoothing
methods
(Error, Trend, Seasonality)



ARIMA

Autoregressive Integrated
Moving Averages

SARIMAX

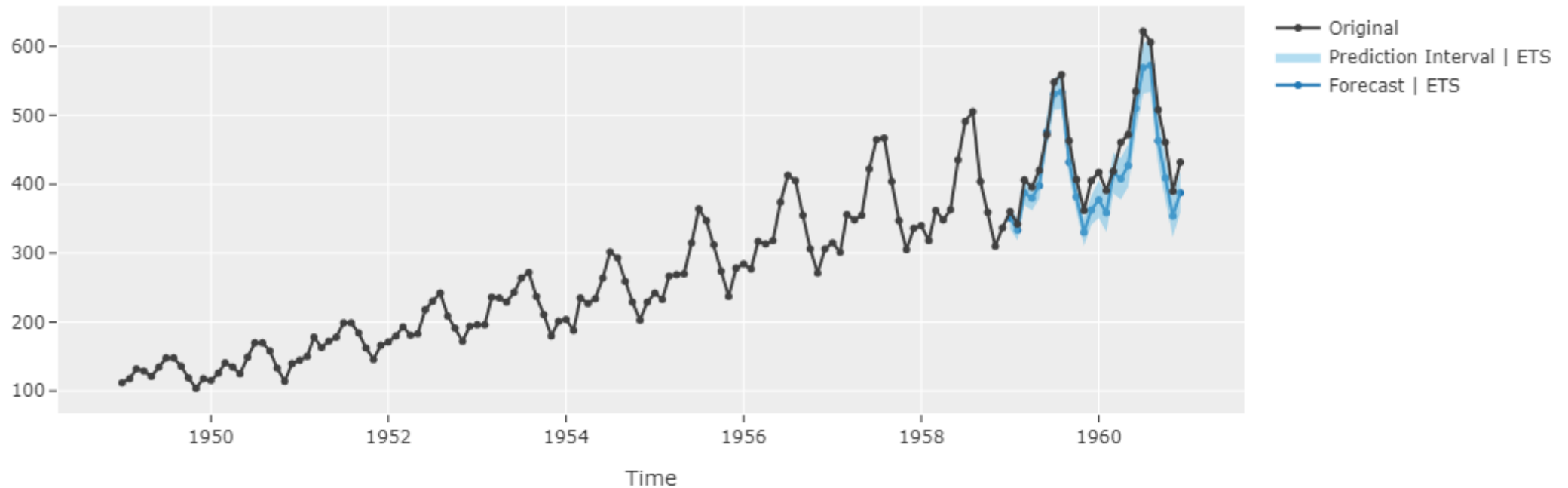


$$y'_t = c + \phi_1 y'_{t-1} + \dots + \phi_p y'_{t-p} + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q} + \varepsilon_t$$

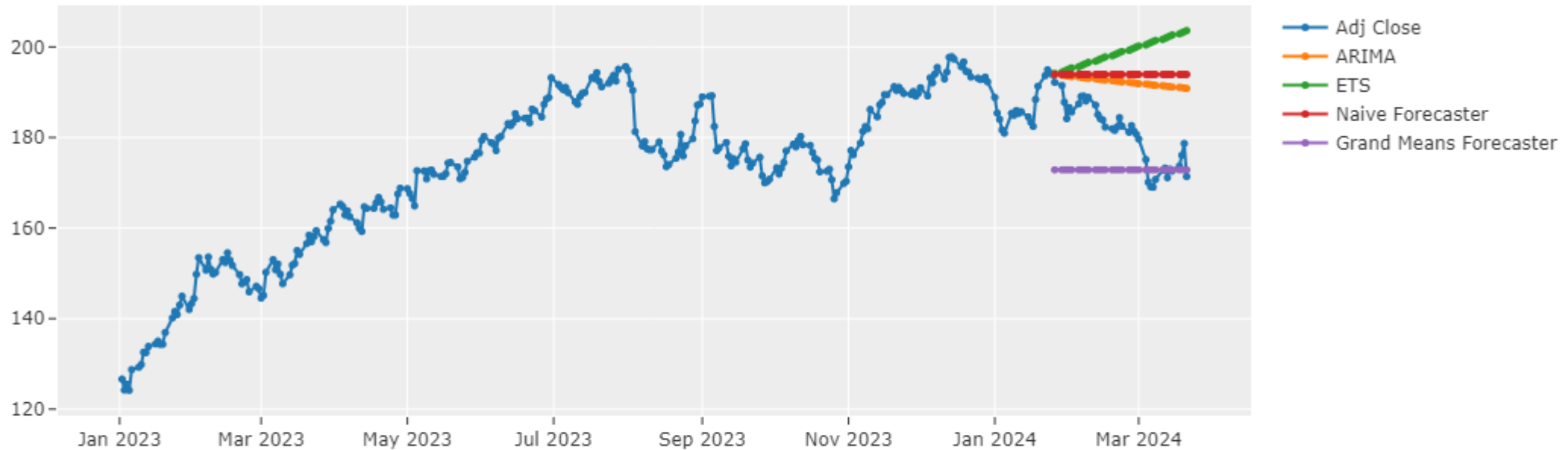


Econometrics Example: ETS

- ETS decomposes time series into Error, Trend, and Seasonal components.



Econometrics Limitations



- Cannot handle strong, complex non-linear relationships
- **Transition to Machine Learning:** ML/DL models are inherently designed to capture complex non-linear patterns in time series

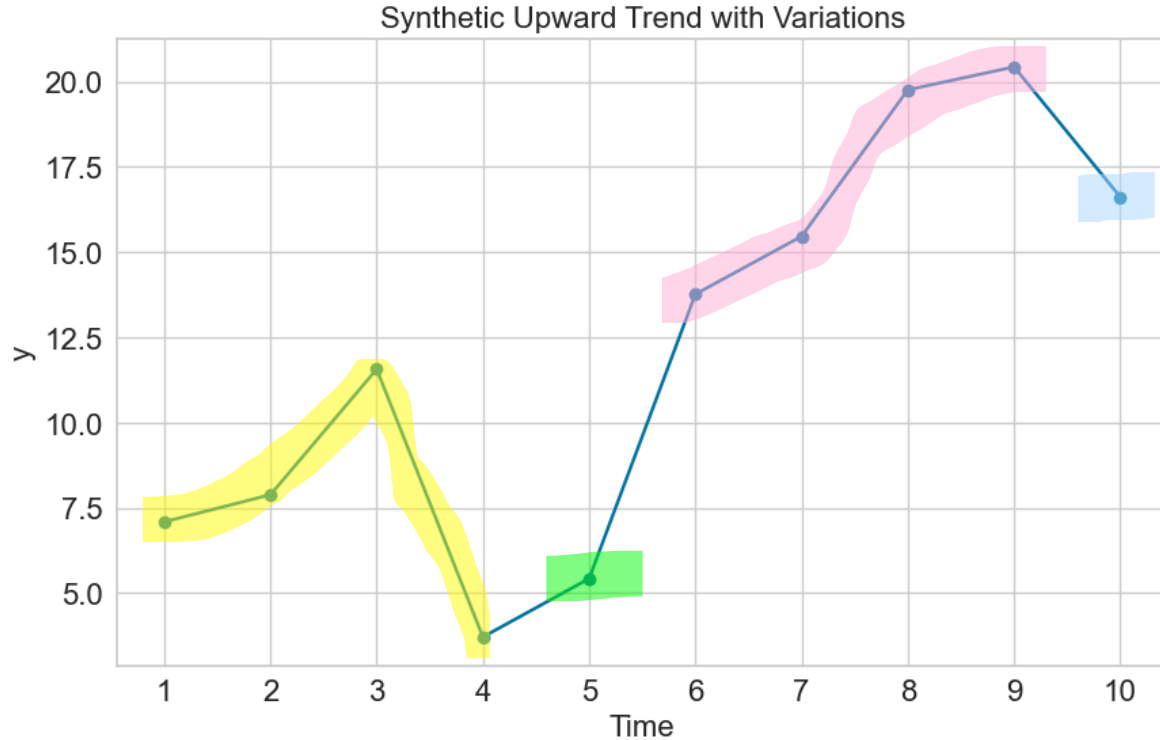


Machine Learning

- **Data Transformation**: preprocessing of time series data into a supervised learning format, creating features and a target variable.
- **Flexibility**: Can capture complex nonlinear relationships without explicit mathematical models for the underlying processes.
- **Feature Importance**: Offers insights into which features (lags, external variables) are most predictive.
- **Challenges**: Overfitting, plus Requires special cross validation and bootstrapping



ML Data Transformation (Single-output)



Features (X)				Target (y)
y_{t-4}	y_{t-3}	y_{t-2}	y_{t-1}	y_t
y_1	y_2	y_3	y_4	y_5
y_2	y_3	y_4	y_5	y_6
y_3	y_4	y_5	y_6	y_7
y_4	y_5	y_6	y_7	y_8
y_5	y_6	y_7	y_8	y_9
y_6	y_7	y_8	y_9	y_{10}

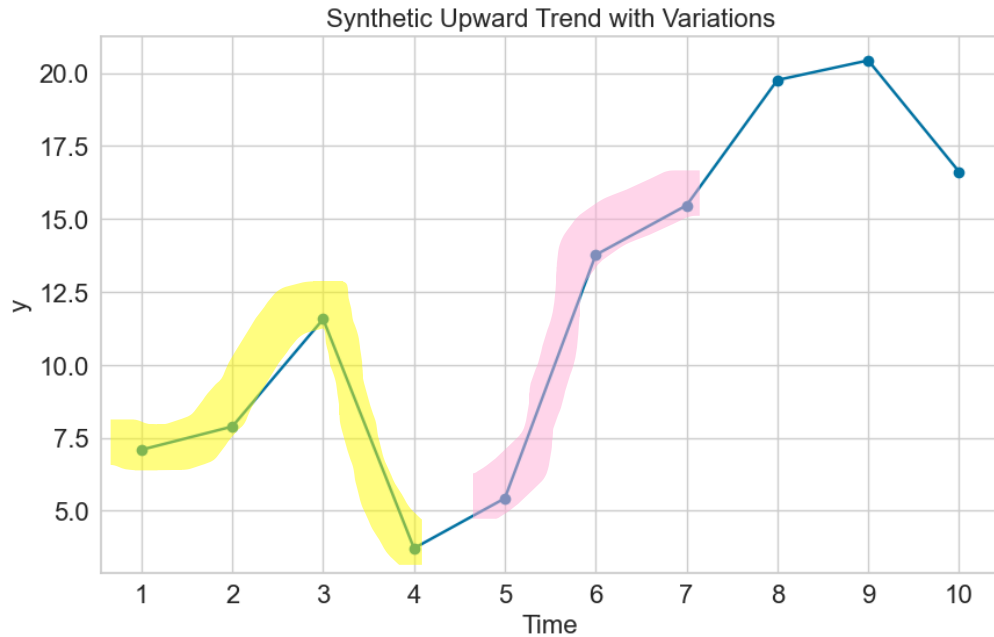
TS raw data

y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9	y_{10}
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→ TS Supervised data



ML Data Transformation (Multi-output)



Features (X)				Target (y)		
y_{t-4}	y_{t-3}	y_{t-2}	y_{t-1}	y_t	y_{t+1}	y_{t+2}
y_1	y_2	y_3	y_4	y_5	y_6	y_7
y_2	y_3	y_4	y_5	y_6	y_7	y_8
y_3	y_4	y_5	y_6	y_7	y_8	y_9
y_4	y_5	y_6	y_7	y_8	y_9	y_{10}

TS raw data

y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9	y_{10}
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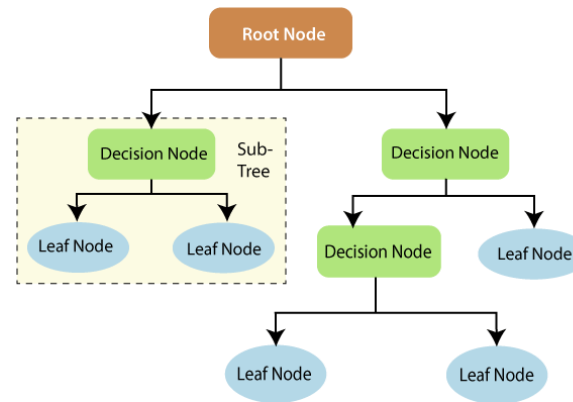
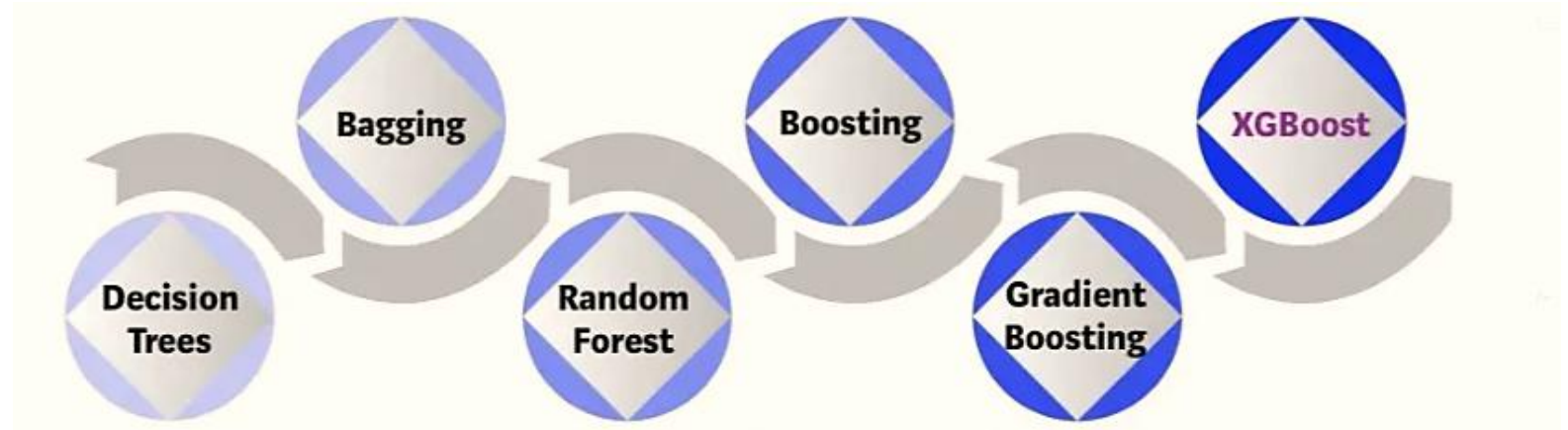
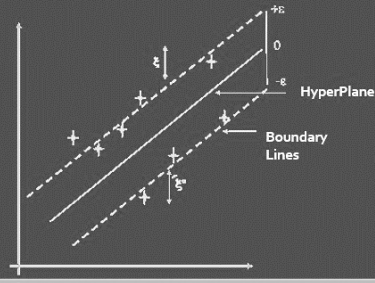


TS Supervised data (h=3)



Machine Learning Models

Support Vector Regression



dmlc
XGBoost

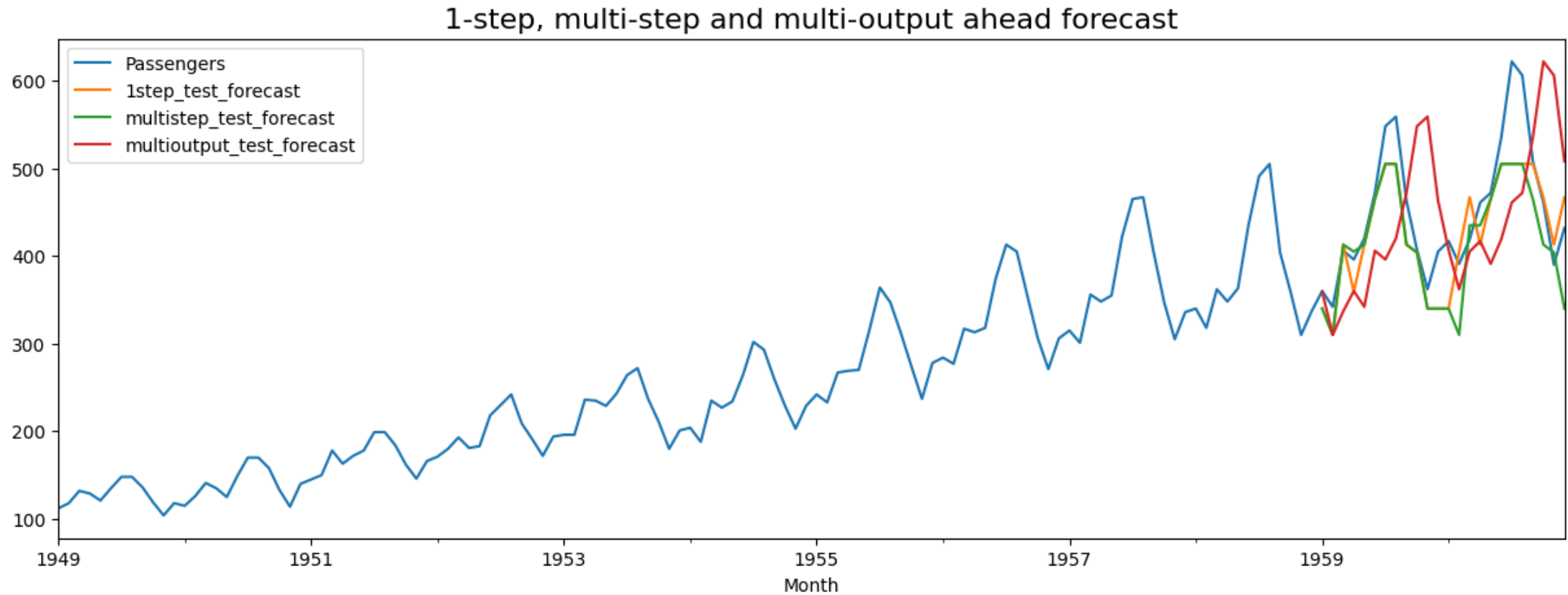
 **LightGBM**

 **CatBoost**



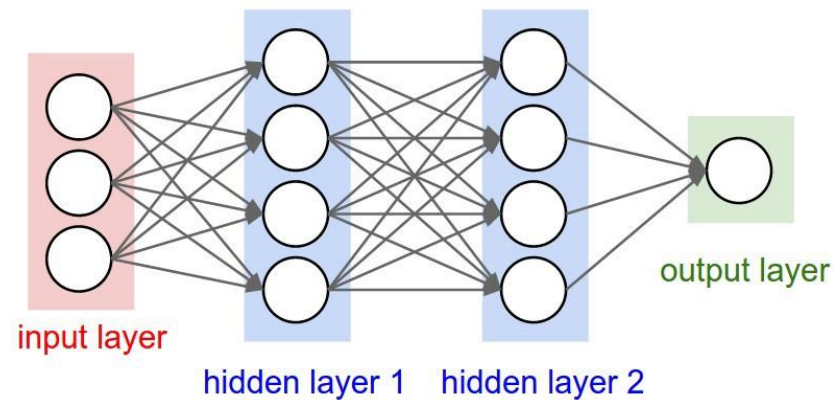
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ML Example: Random Forest with 12 lags

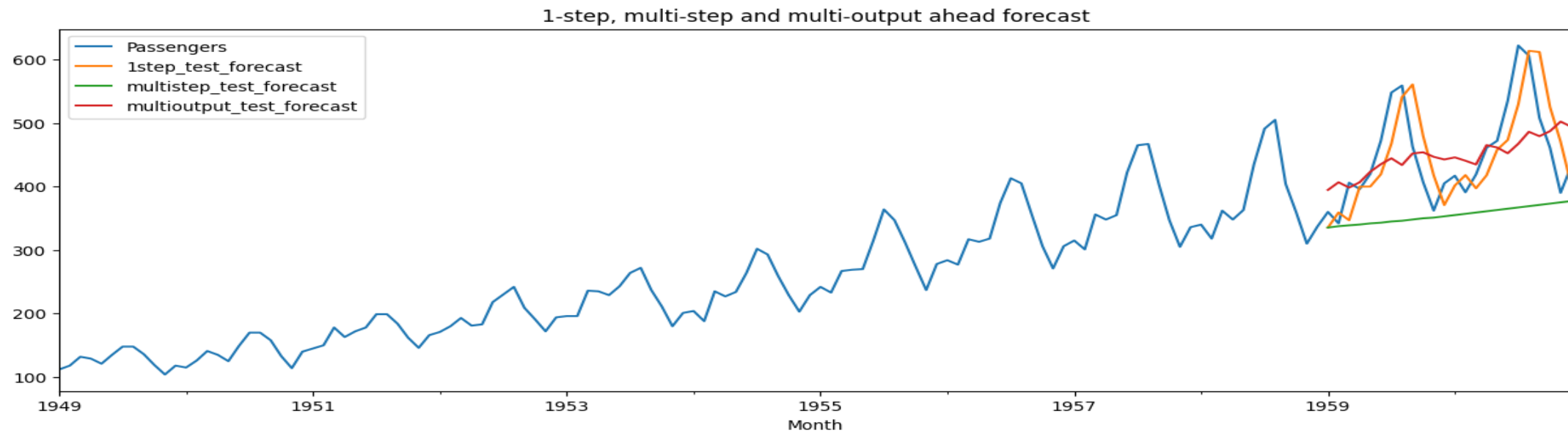
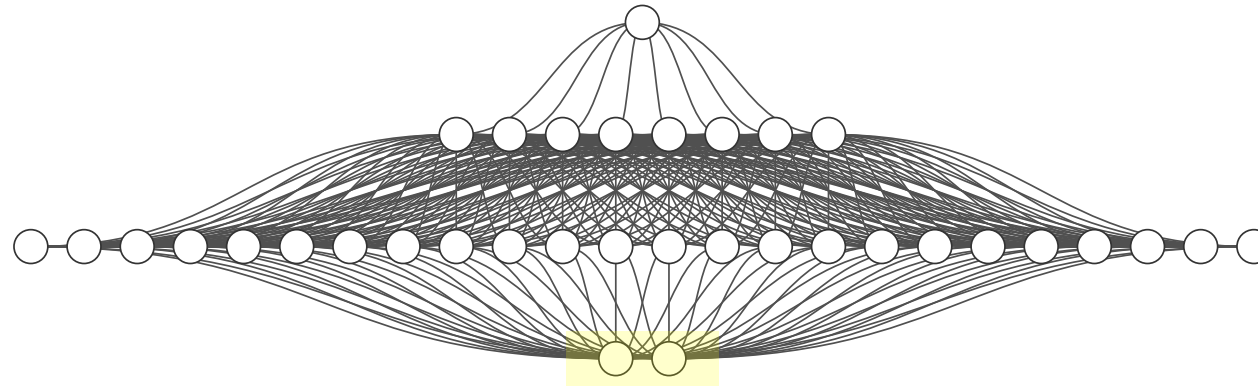


Deep Learning

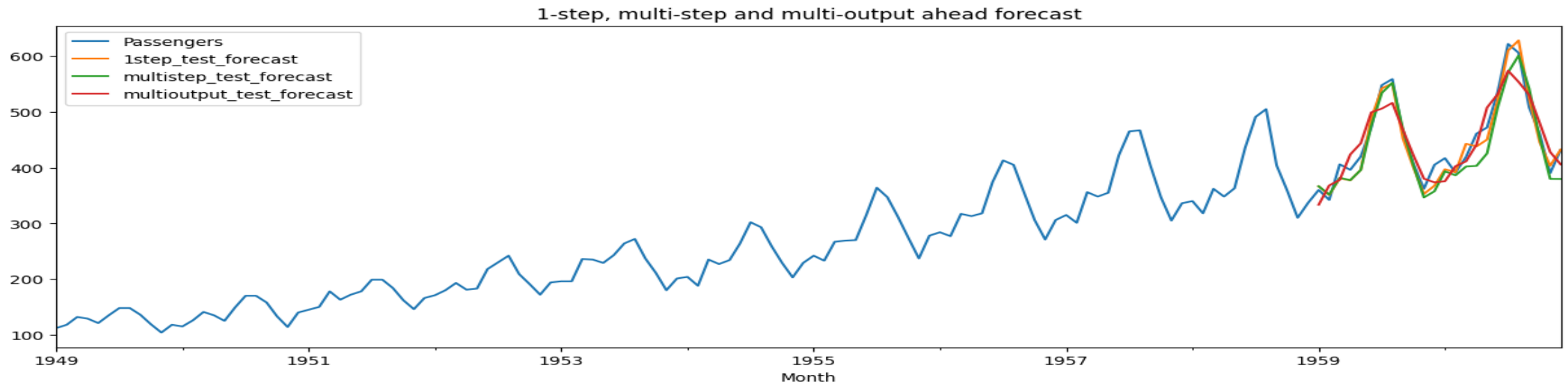
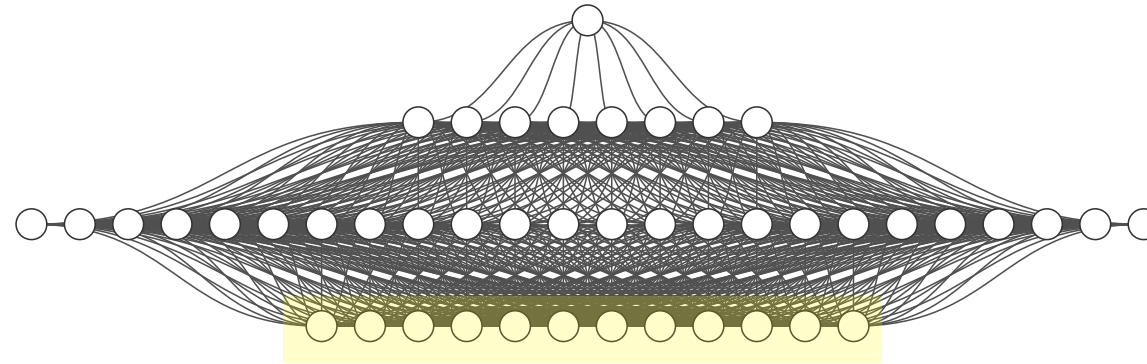
- Just like ML, DL models require Data Transformation.
- **Non-Parametric Nature**: DL models do not assume a predefined form for the time series model, allowing them to learn complex and highly nonlinear patterns directly from data.
- **High Capacity and Flexibility**: Can model very complex relationships and interactions, including multivariate and high-dimensional time series. Well-suited for large datasets.



DL Example: DNN with 2 lags

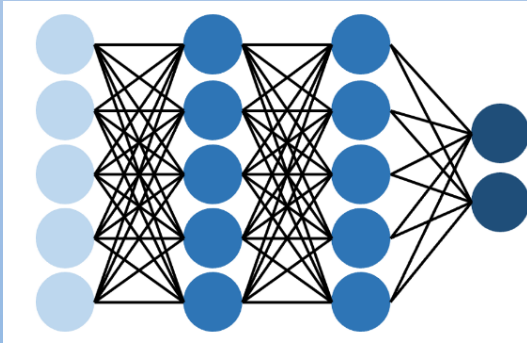


DL Example: DNN with 12 lags

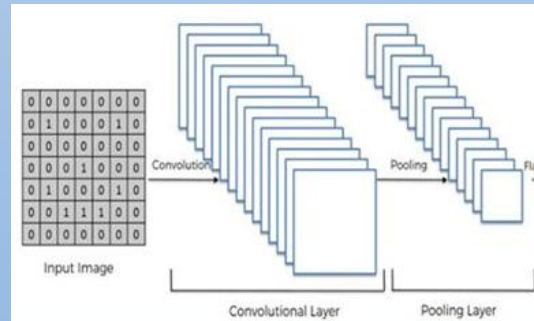


Deep Learning Architectures

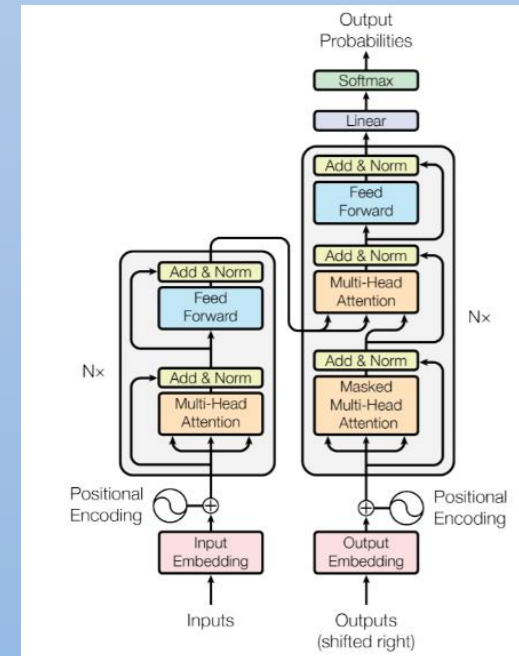
Standard NN



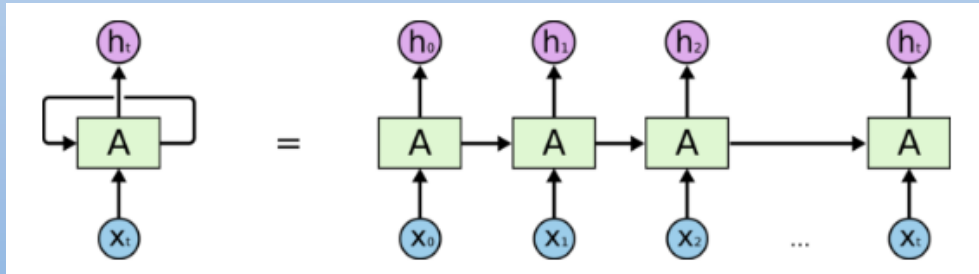
Convolutional NN



Transformers



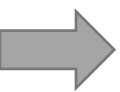
Recurrent NN



Modeling Comparison

Aspect	Econometric Models	Machine Learning (ML)	Deep Learning (DL)
Feature Engineering	Requires explicit modeling of seasonality and trend	Captures complex patterns with less need for manual engineering	Often automates feature engineering
Model Complexity	Lower ; focuses on data generation process	Moderate ; can handle non-linear interactions	High ; suited for complex and high-dimensional data
Interpretability	High ; interpretable parameters, statistical tests	Moderate ; provides feature importance	Low ; considered a 'black box' approach
Computational Intensity	Generally lower	Varies ; dependent on model complexity and data size	High ; requires significant computational resources
Data Suitability	Works well when the underlying process is well understood	Effective for structured datasets with complex relationships	Ideal for large datasets, including unstructured data
Core Models	ARIMA, Exponential Smoothing (ETS)	Random Forest, Gradient Boosting	RNNs, LSTMs, CNNs

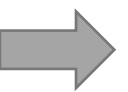




Python packages

Package	Metrics	ML	DL	Focus
 statsmodels	✓✓✓			Statistics, Econometrics
 scikit-learn	✓	✓✓✓		General Machine Learning
 Keras			✓✓✓	General Deep Learning
 PYCARET	✓✓✓	✓✓✓		Auto ML, Rapid prototyping, Comparison
 Darts	✓✓✓	✓	✓✓	Advanced timeseries and forecasting
 GLUONS	✓	✓	✓✓	Large Scale probabilistic Models
 NIXTLA	✓✓✓	✓✓✓	✓✓✓	Workflows, SOTA methods





Module 1- Part 4

1. Timeseries Basics

- What is sequence data?
- Time series tasks
- Trend, Seasonality, Residual
- ACF and PACF
- Stationarity

3. Modeling

- White Noise
- Econometrics
- Machine Learning
- Deep Learning

2. Forecasting strategies

- One-step, Multi-step, Multi-output
- Univariate vs Multivariate
- Benchmarks

4. Can we beat wall street?

- Is stock price Random Walk? (EMH)
- Does Uni-variate Model work?
- Can Multi-variate Models help?
- So What?



Module 1- Part 4

Demystifying Timeseries Data and Modeling



TS Basics
Modeling
Forecasting

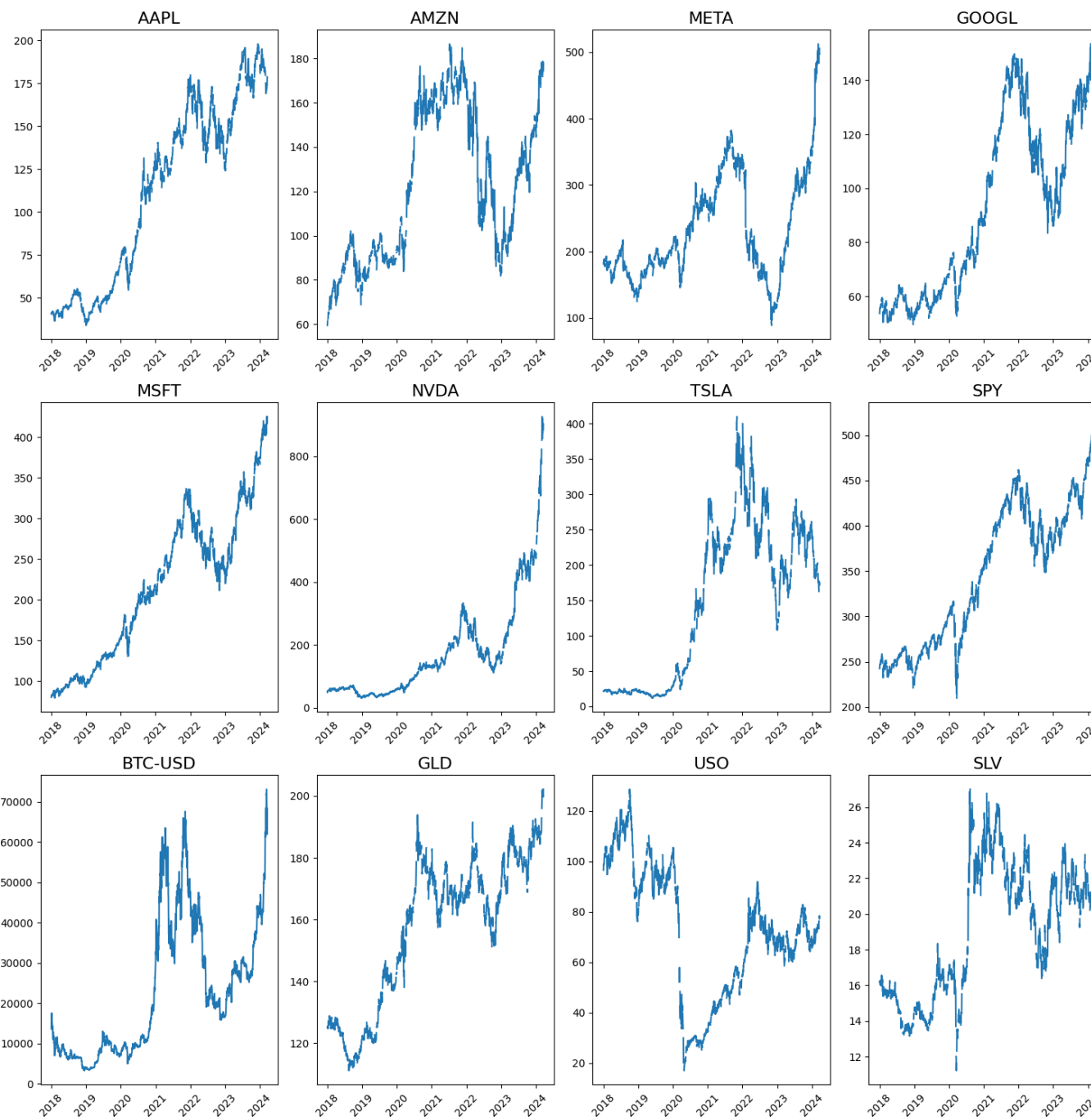


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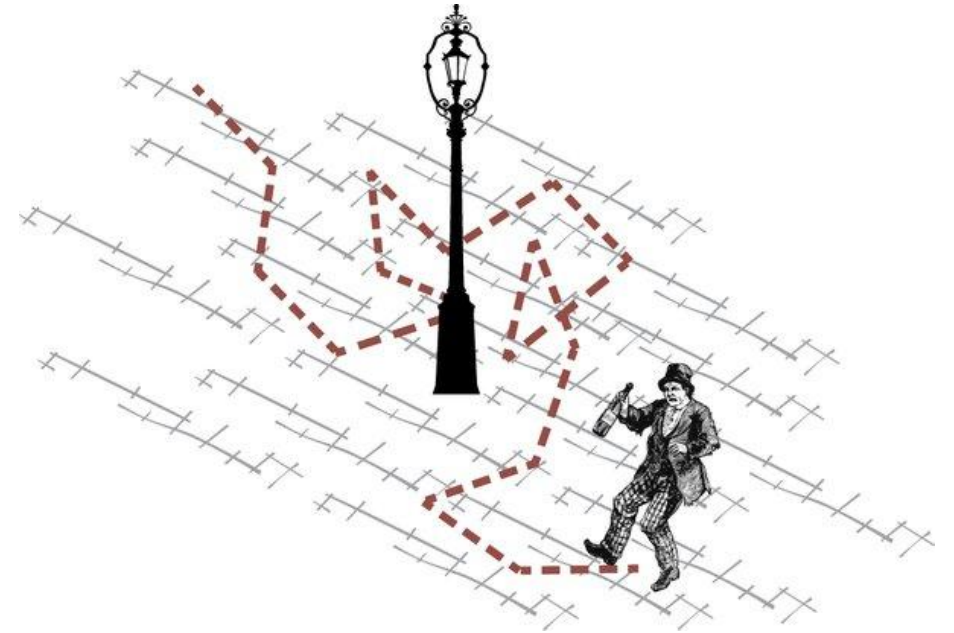
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4. Beating Wall Street



Is stock price Random Walk?

- **Efficient Market Hypothesis (EMH):** Stock prices reflect **all available information**.
- **Empirical evidence:** Statistical tests often fail to reject the random walk hypothesis. Many forecasting models struggle to **consistently outperform naive benchmarks**.
- **Unpredictable Events:** Markets are driven by news, economic shifts, and geopolitical events, many of which are inherently unpredictable and random.
- What are the arguments against random walk?



Is stock price **NOT** Random Walk?

Arguments Against Pure Random Walk

- **Predictability and Patterns:** Evidence of slight predictability and patterns in stock prices, with **technical analysis** exploiting these non-random elements.
- **Behavioral Finance:** Human psychology and **biases**, such as overreaction or herding, introduce **non-random** elements into market behavior.

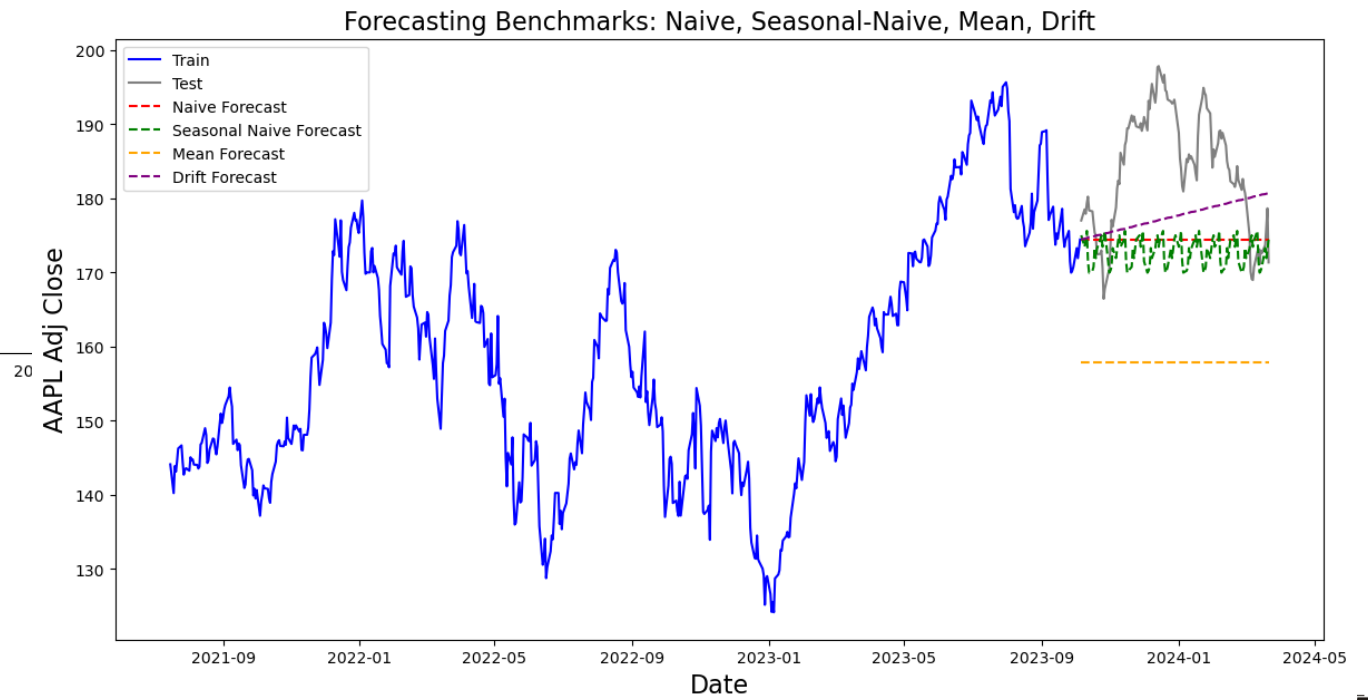
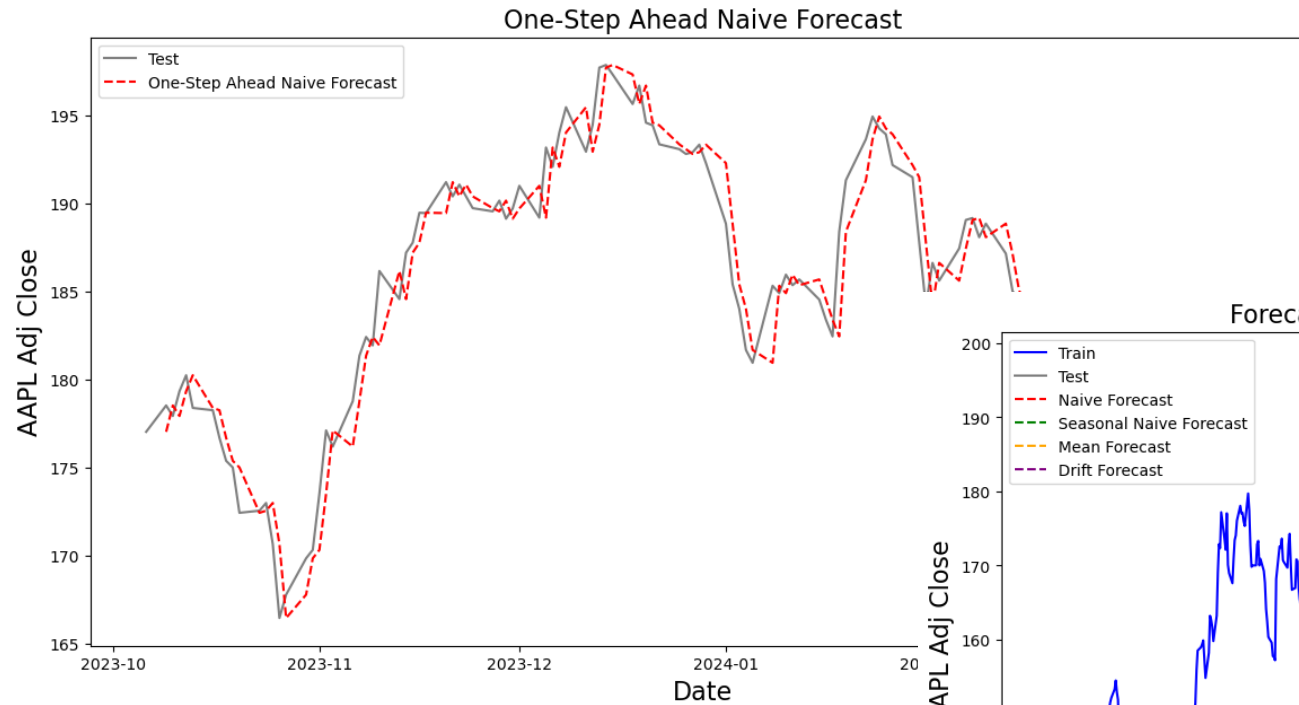


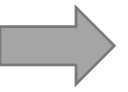
The Middle Ground

- **Not Perfectly Random:** Stock prices likely aren't purely random walks. There may be pockets of predictability or temporary trends.
- **Market efficiency** makes it very difficult to consistently exploit these deviations for significant gains.
- For long-term investors, a random walk model is a **reasonable** approximation →
- **Benchmark = Naïve forecaster**



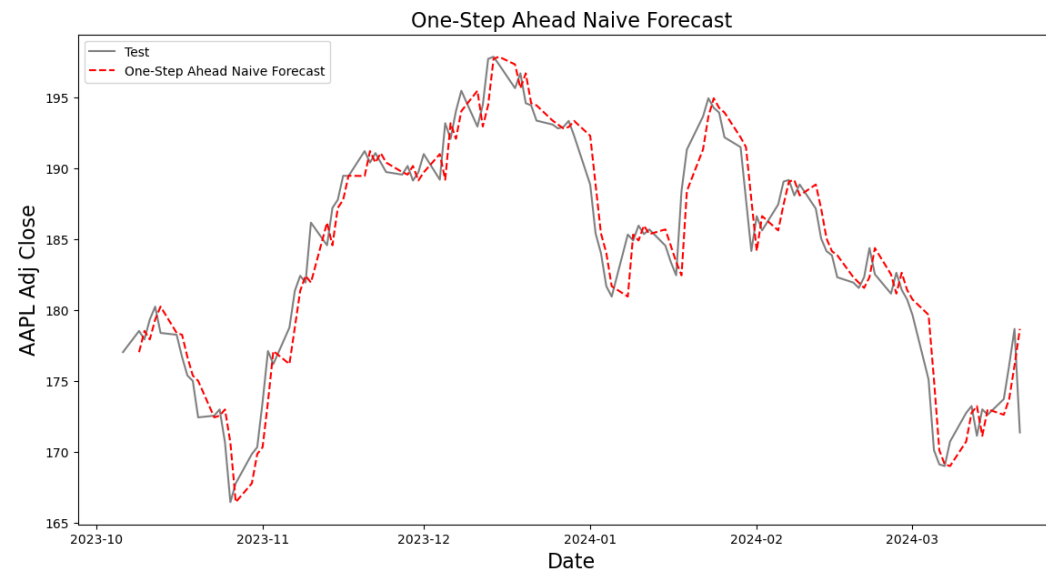
Reminder: Naïve Forecaster

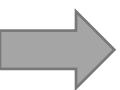




Does Uni-variate Model Work?

- Hard to beat **naïve forecaster!**
- Complex models might offer marginal gains, but the effort/cost of **constant updating** is substantial.
- **Trading strategy** is a whole other challenge: bet sizing, transaction costs, and turning forecasts into profitable decisions.





Can Multi-variate Models Help?

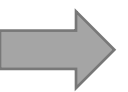
- Potential for Improvement by using **Alternative data**
- **Feature Engineering** is Key: The success of multi-variate models often hinges on identifying and extracting strong, predictive features from the diverse data.
- Increased Complexity and **accessibility challenge**: Building and maintaining multi-variate models is often computationally more demanding
- Trading strategies remains complex due to market efficiency and transaction costs.





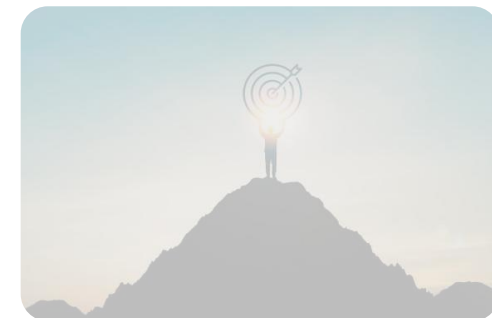


DALL-E



Road map!

- ✓ Module 1- Demystifying Timeseries Data and Modeling
- Module 2- Setting up Deep Forecasting Environment
- Module 3- Exponential Smoothing
- Module 4- ARIMA models
- Module 5- Machine Learning for Time series Forecasting
- Module 6- Deep Neural Networks
- Module 7- Deep Sequence Modeling (RNN, LSTM)
- Module 8- Prophet (Forecasting at Scale)



Pedram Jahangiry

➔ Where to find the materials?



PJalgotrader

Pedram Jahangiry
PJalgotrader

Professor at the Jon M. Huntsman School of Business, Utah State University.

Pinned

 **Machine_Learning-USU** Public

GitHub repository for machine learning course owned and maintained by prof. Jahangiry

Jupyter Notebook ☆ 29 🔗 36

 **Deep_forecasting-USU** Public

GitHub repository for deep forecasting courses owned and maintained by prof. Jahangiry

Jupyter Notebook ☆ 5 🔗 6

 **USU-Analytics-Solution-Center/Bruno.jl** Public

Repository for the Julia package with financial tools for data generation, asset pricing and strategy testing

Julia ☆ 35 🔗 5

 **Deep_Learning-USU** Public

GitHub repository for deep learning courses owned and maintained by prof. Jahangiry

Jupyter Notebook ☆ 7 🔗 6

 **ML-USU-SP2**

ECN 5090- Machine

HTML ☆ 68

 **platforms-and**

GitHub repository for Google Colab, VScode

Jupyter Notebook



You Tube



Pedram Jahangiry

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Professor of data analytics! Machine learning and AI enthusiast 🤖🇺🇸
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